

VR-Sets

An Operational Set Theory Unifying ZFC and ZFA through Reference Semantics

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Abstract

Building on VR (Reznik, 2026, DOI: 10.5281/zenodo.20212092) and VR-Numbers (Reznik, 2026, DOI: 10.5281/zenodo.20272743), we develop *VR-Sets* — an operational theory of sets that unifies ZFC and ZFA through reference semantics. A set is not a container with contents, but an operational functionality: a procedure that, upon a query, either reveals a specific element or reveals nothing. The membership relation $x \in y$ is read as *reference* (the functionality y uses x in its description), not as physical containment. $\emptyset$ is not an ontological primitive but the empty functionality — a nullary operation carrying no ontological content, characterised solely by $\forall x \neg(x \in \emptyset)$; everything else arises as operational action upon it. There are only operations. The closure principle — every operationally describable functionality is a set — replaces the existence axioms of classical ZF and dissolves Russell's paradox operationally: paradoxical descriptions simply do not specify functionalities. All eight ZF axioms together with AC are derived as closure theorems of the operational universe (or follow from the definitions). The theory distinguishes two structural modes: the ZFC-mode (well-founded functionalities) and the ZFA-mode (admitting self-reference and cycles); both are legitimate, and the choice between them is an explicit foundational decision. The Quine atom $A = \{A\}$ and other non-well-founded sets are first-class objects of the ZFA-mode. Three principal divergences from classical ZF emerge: $\wp(\omega)$ is countable; the replacement schema

collapses to a single theorem; foundation is mode-dependent. The VR numbers (von Neumann ordinals) of VR fall automatically into the ZFC-mode, and the entire arithmetic of VR and VR-Numbers is inherited without re-proof via the operational isomorphism. The whole operational universe is countable, which dissolves Cantor's diagonal argument operationally (the diagonal procedure does not specify a describable functionality), makes AC a theorem rather than an axiom, and excludes the Banach–Tarski paradox by construction. Tarski's theorem on the undefinability of truth is reformulated through the hierarchy of applications of the successor operator t . Part X (added in version 1.0.1) documents the Lean 4 formalisation of Parts II–V (Reznik 2026, Lean VR-Sets, DOI: 10.5281/zenodo.20354340), reporting ten methodological observations from the formalisation, centring on five structural boundaries between the operational universe and mathlib's type-theoretic infrastructure — the deepest of which is the constructive (axiom-free) absence of the ZFA-mode in mathlib's `PSet`.

Keywords: foundations of mathematics, operational set theory, ZFC, ZFA, Anti-Foundation Axiom, Quine atom, bisimulation, von Neumann ordinals, countable universe, operational minimalism, closure principle, reference semantics, Lean 4, formal verification, mathlib, axiom audit, structural boundaries.

Part I. Foundations

I.1. Place in the cycle of works

VR-Sets is an operational theory of sets, the third work in a cycle developing the VR programme:

VR. A Formal System (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20212092) — a formal axiomatisation of arithmetic on three primitives $\{\emptyset, \rightarrow, t\}$ and four axioms. It is equivalent to Peano arithmetic and consistent relative to ZF. The basic signature: three primitive operations — the nullary $\emptyset$, implication $\rightarrow$, and the unary successor t (no ontological inventory).

VR-Numbers (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20272743) — operational extensions over the natural numbers of VR. The integers $\mathbb{Z}$, rationals $\mathbb{Q}$, reals $\mathbb{R}$, and complex numbers $\mathbb{C}$ are constructed without introducing ordered pairs as objects. In place of "a pair of numbers" — operational procedures with order built into the description itself. Substantively: $\mathbb{R}$ VR is isomorphic to the field of computable reals (a countable subfield of the classical $\mathbb{R}$); the two-axis structure of $\mathbb{C}$ is structurally motivated by the duality of axiom A1 (the two implications $F \rightarrow F$ and $F \rightarrow \top$).

VR-Sets — the present work. An operational theory of sets that unifies ZFC and ZFA through reference semantics. It resolves the three open questions left in Part VII.3 of VR-Numbers: an operational reformulation of the ZF axioms, the relation to Tarski's theorem on the undefinability of truth, and the status of the axiom of choice.

The common methodological line of the three works is **no ontology at all: $\emptyset$ is itself a nullary operation, with everything else as operational actions**. The slogan, formulated in VR-Numbers Part VI.5: "nothing is — all is doing." In VR-Sets this principle is applied to sets: a set is its own operationality, not a separate entity standing behind it.

I.2. The central idea

In classical ZF, a set is a container. The relation $x \in y$ is read as "x is located inside y." Paradoxes of self-reference (Russell) and paradoxes of choice (Banach–Tarski) arise from tensions in this container metaphor: a container of "all containers that do not contain themselves" contradicts itself; a container holding "all" elements of a continuum turns out to be a source of non-measurable decompositions.

In VR-Sets, a set is a functionality. A set does not "contain" elements; a set is the procedure that, when asked "what are your elements?", either reveals a definite element or reveals nothing. The elements are themselves sets (themselves functionalities). The sole initial entity is $\emptyset$, the empty functionality, which reveals nothing.

The membership relation $x \in y$ is read as reference: x was used in constructing the functionality y. Not as physical containment. This dissolves the paradox of self-reference

at the foundational level: $x \in x$ does not mean "x is physically inside itself" but rather "the functionality x, in one of its responses, returns x." This is a definite response, not a structural cycle.

Two foundational modes correspond to two choices regarding self-reference. The ZFC-mode forbids functionalities from referring to themselves in their responses; the ZFA-mode permits this. In both modes a set is operationality; the difference lies in which operationalities are admitted. Part IV of the preprint formalises this distinction.

Summary of the central idea. A set is not a vessel with contents, but a description of behaviour: "here is what I will reveal when asked." Everything else is consequence of this change of metaphor.

I.3. Ontological position

We fix here the foundational commitments inherited in VR-Sets from VR and VR-Numbers.

Position 1: $\emptyset$ is the base operation, not an ontological primitive

In VR-Sets there is no primary object: $\emptyset$ is the base operation, the empty operationality. All other objects are constructions over $\emptyset$, realised through operational functionalities. No atoms (in the sense of urelements in ZFA), no individuals, no ordered pairs as independent objects. Anything that is "something else" is an *action*, not a separate entity.

Position 2: all is sets

The elements of sets are themselves sets. This is consistent with Position 1: if elements could be some "atoms" distinct from sets, we would have a second primitive kind of object, violating minimalism. In VR-Sets, the only thing that can be revealed upon querying a membership functionality is another set (another functionality, or the same $\emptyset$).

Position 3: operationality is derived, not a kind of being

No ontological status is asserted in VR-Sets — not even for $\emptyset$, which is itself a nullary operation. Operationality is not a derived entity beneath a being: there is no being to derive from. The only asymmetry is structural — $\emptyset$ is the base (nullary) operation, and every other object is built by operations upon it and upon the results of prior operations. There are only operations.

This distinction is structural, not ontological. There are not "two kinds of being — $\emptyset$ and operationality": $\emptyset$ is itself an operation, so there is one kind throughout — operations. The precise formulation: no object is ontologically primary, $\emptyset$ included; the only privilege of $\emptyset$ is syntactic — it is the nullary (base) operation, the one operation taking no prior result, upon which all others act. There are only operations.

A clarifying analogy — corrected as it is drawn. Naively one thinks the number 1 "is" and "add 1" is a mere action; but in VR even 1 is an operation — the successor applied to $\emptyset$. There is no primary thing behind the counting: $\emptyset$ is the nullary operation, succ the step, every number a term over them. So too in VR-Sets: $\emptyset$ is not a primary being but the base operation, and "functionality" is operation itself, not action beneath a being. There are only operations.

No atoms, individuals, labels, urelements, or ordered pairs as objects belong to the operational world of VR-Sets. The name of a set is the set itself (is operationality itself); not a separate syntactic pointer with a status of its own. The slogan "nothing is — all is doing" in application to this position means literally: nothing simply is — not even $\emptyset$; everything is doing, and $\emptyset$ too is an operation.

Position 4: operational infinity in place of actual infinity

Infinity in VR-Sets is the **unbounded applicability of an operation**, not a "completed totality." The set ω is infinite not because it holds all natural numbers as a finished object, but because its functionality is not exhausted at any finite number of queries. The distinction is methodological: classical ZF postulates the existence of an actually infinite set; VR-Sets discovers operational infinity as a consequence of the closure principle (see Part III).

Position 5: operational set in place of set-as-container

A set in VR-Sets is an operational entity given by its membership functionality. A set is its functionality, not separate from it. This is the core definition, given formally in Part II. Here we fix it as a foundational position: "set = operationality" is not a renaming, but a replacement of the underlying metaphor.

The slogan

The slogan of the position, inherited from VR-Numbers Part VI.5 and applied to sets:

"Nothing is — all is doing."

Applied to VR-Sets: a set is its operationality, not separate from it; $\emptyset$ is not a primary object but the base operation; all sets, $\emptyset$ included, are operations, not entities.

I.4. Reference vs. containment: the key distinction

The central distinction on which the resolution of self-reference paradoxes in VR-Sets rests is the distinction between membership as reference and membership as containment. We examine it through simple examples.

The container picture (classical ZF)

In classical ZF the set $\{a, b, c\}$ is a "container" physically holding the elements a, b, c . The self-referential set $A = \{A\}$ on this picture requires that A *contain itself as an element* —

which generates the intuitive paradox: "how can a container be inside itself?" Russell's paradox is a sharpening of this difficulty.

Classical ZF resolves the paradox syntactically (axiom of separation: one cannot form a set from elements with an arbitrary property, only from elements of an already existing set with a property); type theory resolves it by stratification (a set and its element must lie at different levels). Both solutions preserve the container metaphor while imposing restrictions on it.

The reference picture (VR-Sets)

In VR-Sets a set is not a container. A set is a procedure that answers queries. $x \in y$ means: "the functionality y , in one of its responses, reveals x ." This is *reference*: y uses x in its description.

The self-referential set $A = \{A\}$ on this picture: A is a functionality that, upon a query, reveals A . There is no "physical embedding of itself within itself": there is a procedure that returns a definite response. That this response happens to be A is no more paradoxical than a recursive function calling itself within its own definition. The structure is familiar to a programmer:

```
function A() { return A; }
```

— there is no contradiction here. A is a function whose only response is itself. This is an operationally defined description.

Where the difference lies

The distinction between the two pictures is not mathematical (the mathematics of the same sets remains the same) but **semantic**. The container picture reads $x \in y$ as a claim about spatial structure; the reference picture reads it as a claim about computational behaviour. If this were merely a renaming, the distinction would be empty. But it is substantive:

(a) **Russell's paradox** in the container picture requires syntactic restrictions (axiom of separation). In the reference picture it is dissolved operationally: the description $R = \text{"reveal } x \text{ if and only if } x \notin x\text{"}$ demands a contradictory response to the query " $R \in R$?" and therefore does not specify a describable functionality. R is not "forbidden" — it *does not exist* as a functionality.

(b) **Self-membership** in the container picture is a pathology, requiring a separate axiom (anti-foundation) to be admitted. In the reference picture it is a natural possibility, restricted only by the structural condition of well-foundedness (Part IV). Whether to admit it or forbid it is an foundational choice, not a mathematical necessity.

(c) **Extensionality** in the container picture is a separate axiom (sets with the same elements are equal). In the reference picture it is a tautology of the identity definition: sets with the same responses to queries are the same functionality (see Definition 4 of Part II).

An analogy from programming

The distinction between reference and containment is routine in programming. Given a structure

node = { value: 1, next: node }

— this is a cyclic linked list. There is no "infinite embedding" in memory; there is a pointer node.next referring to node. The structure is well-defined, not pathological, not paradoxical. The programmer distinguishes "node holds next as a field" (a reference) from "node physically contains a copy of itself" (which would be an infinite matryoshka). VR-Sets makes the same distinction for sets: $A \in A$ is a pointer from the functionality A to itself, not an infinite matryoshka.

This analogy is substantive, not decorative. Modern approaches to non-well-founded sets (Aczel 1988, Barwise–Moss 1996) explicitly exploit graph representations in which cycles are admissible data structures. VR-Sets does the same at the foundational level.

I.5. Countability of the operational universe

A fundamental consequence of the operational universe: the entire universe of sets in VR-Sets is countable. This is a sharp divergence from classical ZF, where the universe V is a proper class of uncountable cardinality.

The argument

Every set in VR-Sets is given by a *description* of its functionality. A description is a finite syntactic structure (even when the functionality reveals countably many elements — as with ω — the description itself is finite: "the n -th query yields O_n "). The set of finite descriptions over a finite alphabet is countable: this is a basic fact of the theory of formal languages (a countable union of finite sets is countable).

Therefore the operationally describable functionalities are countable in number. By the closure principle (Part II §5), every describable functionality is a set. Hence the sets in VR-Sets are countable in number.

Comparison with classical ZF

In classical ZF the sets form a proper class of uncountable cardinality, and $\wp(\omega)$ alone is already uncountable (Cantor's theorem). VR-Sets departs radically from this picture: $\wp(\omega)$ is countable in VR-Sets (Part III, §III.5), and this is not a defect but a consequence of the foundational choice.

Detailed discussion of this divergence is given in Part VI. Here we fix it as a fundamental fact: **the operational universe of VR-Sets is countable, and from this follow many structural features of the theory** — the absence of uncountability, the simplification of the axiom of choice, the absence of the Banach–Tarski paradox.

I.6. Terminological glossary

We fix here the terminology used consistently throughout all parts of the preprint.

Ontological terms

Operational set — a set in VR-Sets, given by its membership functionality.

Operationality — the membership functionality, that which is the set. Used as a synonym for "functionality" in contexts where the status of an object is at issue.

Functionality membership (or simply functionality) — a procedure responding to queries "what are your elements?"; either reveals a definite element or reveals nothing.

Operational infinity — unbounded applicability of an operation, contrasted with the actual infinity of classical ZF.

Operational depth — the least number of unfolding steps after which a functionality reaches $\emptyset$. The depth of $\emptyset$ is 0; finite sets have finite depth; ω has non-finite but non-cyclic depth; cyclic sets (such as the Quine atom) have no well-defined depth.

Operational identity $\equiv$ — coincidence of functionalities: $A \equiv B$ if every element of A matches an element of B under $\equiv$, and conversely. The analogue of Leibnizian equality from VR.

A note on the sign $\equiv$

The sign $\equiv$ in VR-Sets denotes operational identity. It is **not** used in any of the standard alternative senses that this sign carries in mathematical literature:

- not congruence modulo n (as in number theory);
- not logical equivalence (as between formulas);
- not definitional equality (as in type theory).

Rather, $\equiv$ is the coinductive identity between operational functionalities, defined in Part II Definition 4. It is structurally equivalent to bisimulation in the AFA literature (Aczel 1988), and readers familiar with that tradition may regard $\equiv$ as bisimulation lifted to the operational setting. We use $\equiv$ rather than $=$ to signal explicitly that identity of sets in VR-Sets is defined through coincidence of functionalities, not through any pre-existing notion of equality.

Syntactic terms

Description of a functionality — a finite syntactic record specifying an operational procedure. A set is describable if it has a description.

Name of a set — the set itself. The operational world of VR-Sets contains no separate layer of "names denoting objects": the name and the named coincide.

Mode-related terms

ZFC-mode — the operational universe restricted to well-founded functionalities. Formally defined in Part IV.

ZFA-mode — the operational universe without the well-foundedness restriction; admitting self-reference and cycles.

Well-founded — a functionality is well-founded if every sequence of successive unfoldings of its responses terminates at $\emptyset$ in a finite number of steps. Discussed in detail in Part IV §IV.2.

Meta-principles

The closure principle — every operationally describable functionality is a set. Formal statement in Part II §3. In ZFC-mode the principle is applied subject to well-foundedness; in ZFA-mode without restriction.

Operational definiteness — a functionality returns a definite response to every query; contradictory descriptions (requiring contradictory responses) do not specify functionalities. This is what dissolves Russell's paradox.

I.7. Structure of the preprint

The preprint consists of ten parts, listed below. The present part is the first.

Part I. Foundations. Place in the cycle of works; the central idea; foundational position; the distinction between reference and containment; countability of the operational universe; terminological glossary; structure of the preprint.

Part II. The Operational Construction of Sets. Four core definitions (set, $\emptyset$, kinds of sets by operability, operational identity $\equiv$), three lemmas (extensionality, uniqueness of $\emptyset$, operational depth), the closure principle, the resolution of Russell's paradox, principal examples.

Part III. The ZF Axioms as Closure Theorems. All eight ZF axioms together with AC are closure theorems of the operational universe (or follow from the definitions of Part II). Divergences from classical ZF: $\wp(\omega)$ is countable; the replacement schema becomes a single theorem; foundation is a mode-dependent property.

Part IV. The Two Modes: ZFC and ZFA. Formal definition of the ZFC-mode and ZFA-mode through well-foundedness as a structural property of a functionality. Theorems of completeness of each mode relative to the corresponding classical theory. Conjecture of equivalence — via a countable model.

Part V. The Numbers of VR in VR-Sets. The von Neumann ordinals On and ω as specific well-founded operational sets. Automatic membership in the ZFC-mode as a theorem. Arithmetic inherited via isomorphism with VR Part I.

Part VI. Cardinality, Cantor, Choice. Countability of the operational universe in full detail. Cantor's diagonal argument in the operational universe. AC as a theorem on the countable universe. Absence of the Banach–Tarski paradox by construction.

Part VII. Truth and Self-Reference. Tarski's theorem on the undefinability of truth in light of the operational universe. The duality of axiom A1 and the hierarchy of levels of truth through applications of t . A cautious formulation: a reformulation, not a resolution.

Part VIII. Ontological Position. A parallel to VR-Numbers Part VI. "Nothing is — all is doing" applied to sets, with detailed exposition of all foundational commitments.

Part IX. Open Questions. Connections with topoi, homotopy type theory (HoTT), formalisation in Lean, machine verification of equivalence conjectures, extensions (Morse–Kelley, NBG).

Part X. Lean 4 Formalisation of VR-Sets: Methodological Observations.

Methodological observations arising from the Lean 4 formalisation of Parts II–V (Reznik 2026, Lean VR-Sets, DOI 10.5281/zenodo.20354340; tag v1.2-vr-sets). Ten observations in four groups: quotient-base structure (A); five structural boundaries between the operational universe and mathlib (B), centring on the type-theoretic absence of the ZFA-mode; axiom-minimal patterns (C); methodological convergences (D). Resolves §IX.1 Question 3 of this preprint for Parts II–V.

Each part can be read with minimal context from preceding ones, but Parts II–IV form the core of the system, on which the remaining parts rest.

Part II. The Operational Construction of Sets

II.0. Introduction

This part of the preprint formalises the operational construction of sets. Building on the position fixed in Part I ($\emptyset$ as the base operation, not a primitive; everything else as operationality), we introduce four definitions forming the core of the system:

- **Definition 1** — what a set is in VR-Sets.
- **Definition 2** — what $\emptyset$ is.
- **Definition 3** — three kinds of sets distinguished by the character of their operationality (finite, infinite, cyclic).
- **Definition 4** — operational identity $\equiv$.

From these four definitions three lemmas are derived (extensionality, uniqueness of $\emptyset$, operational depth); the closure principle is stated — the principal instrument by which the ZF axioms will be proved as theorems in Part III; and Russell's paradox is dissolved operationally. The part closes with a set of characteristic examples.

The presentation is structured so that all of Parts III and IV rest only on this text; nothing from the later parts is presupposed. This part is the self-contained foundation.

II.1. Core definitions

Four definitions forming the core of the operational universe of sets. They are fixed in their final form — changes here would affect all that follows.

Definition 1 (set)

A **set** in VR-Sets is an operational entity given by its membership functionality: upon a query, the functionality either reveals an element (which is itself a set, possibly the same set as the one being queried) or reveals nothing. The order in which elements are revealed is an artefact of observation and does not enter into the identity of the set. A set has no internal structure beyond its functionality; it *is* its functionality.

Commentary. (1) A set in VR-Sets is not a container with contents, but a description of behaviour: "here is what I will reveal when asked." This replacement of the underlying metaphor is the core divergence from classical ZF (see Part I §I.4).

(2) "A set is its functionality" is a strict identity, not an operationalism by definition. The functionality is not "how the set behaves" (as though something more fundamental stood behind the behaviour); the functionality is the set. No "carrier of the functionality" distinct from it is presupposed.

(3) The elements of sets are themselves sets. This is explicitly fixed in the definition ("an element is itself a set") and accords with Position 2 from §I.3: in VR-Sets there are no atoms, individuals, or urelements.

(4) Self-reference ("the same set as the one being queried") is explicitly admitted in the definition. This is an operational possibility; restrictions on self-reference pertain to the choice of mode (Part IV), not to the definition of a set as such.

(5) The order of revelation is an artefact of observation. This means that if a functionality reveals elements a, b in one observed order and b, a in another, both observations relate to the same functionality. The identity of a set is determined by *which* elements are revealed, not *in what order*.

Definition 2 ($\emptyset$)

$\emptyset$ is the unique set whose functionality reveals nothing: $\emptyset$ is the empty operationality.

Commentary. (1) The definition appeals to "uniqueness," which will be proved as Lemma 2. Here uniqueness is fixed as part of the meaning of the symbol $\emptyset$: when we write $\emptyset$, we mean precisely this set, not one among many possible empty sets.

(2) $\emptyset$ is the base operation of VR-Sets, the one operation taking no prior result; no ontological primacy is asserted, for it or anything else. All other sets are operationalities; $\emptyset$ is empty operationality, and this "empty" is the terminating point of every construction.

Definition 3 (kinds of sets by operationality)

A set A is:

- *finite* if its functionality exhausts all elements in a finite number of queries;
- *infinite* if its functionality is not exhausted at any finite number of queries;
- *cyclic* if among the revealed elements its functionality returns A itself.

These properties are not mutually exclusive.

Commentary. (1) Finiteness is defined through exhaustion: $\emptyset$ and all the von Neumann ordinals O_n are finite. Every element of any finite set has finite operational depth (Lemma 3).

(2) Infinity is defined through *non-exhaustion*, not through cardinality. Example: ω is infinite — its functionality reveals $O_0, O_1, O_2, \dots$ without termination.

(3) Cyclicity is defined through self-return. The Quine atom $A = \{A\}$ is cyclic. The set $B = \{\emptyset, B\}$ is cyclic (its functionality returns B among its other elements). Cyclic sets belong to the ZFA-mode but not the ZFC-mode.

(4) The properties are not mutually exclusive: a set may be simultaneously infinite and cyclic (for instance, a describable sequence of sets each containing a reference to the

next and to itself). Such cases are rare in substantive mathematics but are not excluded by the definition.

Definition 4 (operational identity $\equiv$)

Sets A and B are *operationally identical* ($A \equiv B$) if their functionalities coincide: for every element x of A there exists an element y of B with $x \equiv y$, and conversely. $\emptyset \equiv \emptyset$.

Commentary. (1) The definition is coinductive: $\equiv$ is defined through $\equiv$ on elements. The termination point is $\emptyset$: if both sets are empty, $\equiv$ holds trivially (vacuously).

(2) Relation to Leibnizian equality from VR. In VR (Part I, Definition 2) equality is given classically: $x = y \iff \forall p: p(x) \iff p(y)$ — coincidence under all properties. In VR-Sets $\equiv$ is the direct analogue of Leibniz, lifted to the operational level: in place of "under all properties" — "under all responses to queries." Substantively, this is the same identity relation, expressed in the language suited to the operational reading.

(3) The distinction between $\equiv$ and ordinary $=$. In what follows $\equiv$ is used for the operational identity of sets, and $=$ only in contexts referring to classical objects (ZF formulas, numerical equalities). This discipline is observed in all parts of the preprint. The sign $\equiv$ is also distinguished from its various standard senses in mathematical literature (modular congruence, logical equivalence, definitional equality); see Part I §I.6 for the note on notation.

(4) Bisimulation. In the AFA literature (Aczel 1988) the analogous relation is known as bisimulation: sets are equivalent if their graphs of membership are isomorphic (allowing cycles). In VR-Sets $\equiv$ is structurally the same relation, expressed operationally. The direct connection is discussed in Part IV §IV.8.

II.2. Lemmas

Three consequences embedded in the definitions. They are used as lemmas in the later parts.

Lemma 1 (extensionality)

If A and B have the same elements (up to $\equiv$), then $A \equiv B$.

Proof. This is a direct reading of Definition 4: the condition of coincidence of functionalities in Definition 4 is precisely "for every element x of A there exists y of B with $x \equiv y$, and conversely." The statement "sets with the same elements are distinct" contradicts the definition of identity and is inexpressible in VR-Sets.

■

Remark. This is not a theorem in the strict sense but a formal restatement of the definition. In classical ZF extensionality is a separate axiom, because in the classical approach equality is given independently of membership. In VR-Sets equality (more

precisely, operational identity $\equiv$) is defined *through* membership, so extensionality is built into the definition.

Lemma 2 (uniqueness of $\emptyset$)

There exists exactly one set with an empty functionality.

Proof. If two functionalities both reveal nothing, the coincidence condition of Definition 4 holds vacuously on both sides: "for every element x of A there exists y of B with $x \equiv y$ " — the premise is false, the implication is true; and conversely. Therefore $A \equiv B$.

■

Remark. Lemma 2 does not assert the existence of $\emptyset$ — this is given by Definition 2 as part of the construction. The lemma asserts uniqueness: in writing $\emptyset$ we refer to a fully determined object, not to an arbitrary "empty set" among many.

Lemma 3 (operational depth)

Every set has an operational depth — the least number of unfolding steps after which $\emptyset$ is reached. The depth of $\emptyset$ is 0. Finite sets have finite depth. Infinite and cyclic sets do not have finite depth (for cyclic sets, unfolding does not terminate at $\emptyset$).

Note on the status of the lemma. Lemma 3 is not an existence theorem but a definition of a property of a functionality. Every set A *has* its depth: a number (or "infinity" with the qualification "non-cyclic") — a structural characteristic of A . For finite sets this number is finite; for ω this number is non-finite but each concrete unfolding terminates (see Part IV §IV.2); for cyclic sets unfolding does not terminate at all.

Significance of Lemma 3. This lemma provides the instrument of *induction on sets* in VR-Sets — the analogue of axiom A4 from VR, lifted to the level of sets. The numbers of VR (the von Neumann ordinals On) have operational depth exactly n ; ω has non-finite depth but is non-cyclic — this is essential for distinguishing "infinite by enumeration" from "cyclic."

II.3. The closure principle

The principal instrument of VR-Sets. Through it all the ZF axioms are proved in Part III.

Statement

The closure principle of VR-Sets. If a functionality F is operationally describable — that is, if there exists an operational procedure that gives a definite response to every query (either revealing a specific element or revealing nothing) — then there exists a set A such that A is F .

A description requiring a functionality to give contradictory responses to one and the same query (as in the definition $R = \text{"reveal } A \text{ if and only if } A \text{ is not revealed"}$) **does not** specify an operationally describable functionality, and does not give rise to a set.

Commentary

(1) The closure principle converts the axioms of set existence (as they appear in ZF) into *closure theorems* of the operational universe relative to specific describable constructions. There is no need to postulate " $\cup A$ exists" — one must show that $\cup A$ is operationally describable, and then by the closure principle it is a set. This is a substantive simplification: in place of a long list of existence axioms, a single general principle.

(2) The closure principle is neither a definition nor a theorem, but an *foundational choice*: which functionalities count as sets. The choice is motivated by the slogan "nothing is — all is doing": all that exists as describable action exists as a set; no different status is presupposed for "potential" or "inaccessible" entities.

(3) "Operational describability" is a substantive notion. A description must: (a) be finite (a syntactic record in a formal language); (b) give a definite response to every query (not contradictory, not undefined, not undecidable by construction). When these conditions hold, the description specifies a functionality, and the closure principle applies.

(4) In the two modes of VR-Sets the closure principle is applied differently. In the ZFC-mode — only to functionalities whose successive unfoldings terminate at $\emptyset$. In the ZFA-mode — to all describable functionalities. Part IV unfolds this distinction.

Resolution of Russell's paradox

Russell's paradox is dissolved at the level of the definition. The description $R = \text{"reveal } x \text{ if and only if } x \notin x\text{"}$ demands a contradictory response from the functionality to the query " $R \in R?$ ":

- If the functionality R reveals R on this query, then by the description (R is to contain only x with $x \notin x$) we have $R \notin R$, so R *should not* reveal R . Contradiction.
- If R does not reveal R , then by the description R should contain R (since $R \notin R$). So R *should* reveal R . Contradiction.

This is a contradictory description. By the closure principle R is **not** a set — not because we forbade it by a separation schema (as in ZF) or by type stratification (as in type theory), but because such a functionality *does not exist* operationally. At each concrete step a functionality either reveals a specific element or does not; no third option exists; the contradictory definition of R cannot be carried out by any procedure.

Relation to classical approaches

Classical ZF resolves Russell's paradox by the **axiom of separation**: one cannot form a set from all x with an arbitrary property, only from $x \in A$ with a property. This is a syntactic restriction imposed on the schema of construction.

Type theory resolves the paradox by **stratification**: an element and a set must lie at different levels; " $x \in x$ " is a type violation.

VR-Sets resolves the paradox by **operational definiteness**: paradoxical descriptions simply do not specify functionalities. The result is the same (no paradox), but the cause is operational, not syntactic. This is consistent with the general programme of VR: operational clarity is placed ahead of technical devices.

II.4. Principal examples

Specific operational sets illustrating the definitions and lemmas.

Example 1: $\emptyset$

The functionality $\emptyset$ reveals nothing. Operational depth — 0. Finite (exhausted in 0 queries), not infinite, not cyclic. Unique by Lemma 2.

Example 2: $\{\emptyset\}$

The functionality $\{\emptyset\}$ reveals $\emptyset$, and on all subsequent queries reveals nothing. Operational depth — 1 (one unfolding step suffices to reach $\emptyset$). Finite.

In the terminology of VR this is $O1$ — the first non-zero von Neumann ordinal, obtained by applying t to $\emptyset$: $O1 = t(\emptyset) = \emptyset \cup \{\emptyset\} = \{\emptyset\}$.

Example 3: $\{\emptyset, \{\emptyset\}\}$

The functionality reveals $\emptyset$ and $\{\emptyset\}$. Operational depth — 2. Finite.

In the terminology of VR this is $O2 = t(O1) = O1 \cup \{O1\} = \{\emptyset, \{\emptyset\}\}$.

Example 4: ω

The functionality ω reveals on its n -th query the von Neumann ordinal O_n : $\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \dots$, without termination.

Operational depth — non-finite (unfolding does not reach $\emptyset$ at any finite step). Infinite (the functionality is not exhausted). **Non-cyclic**: ω does not return itself among the revealed elements. Each revealed element is a specific O_n with finite n ; the unfolding chain from any specific O_n terminates at $\emptyset$ in n steps.

This last characteristic — "infinite but non-cyclic" — gives ω the property of well-foundedness, unfolded in Part IV §IV.2: every successive unfolding of ω terminates at $\emptyset$, even though ω itself has no finite operational depth.

Example 5: the Quine atom $A = \{A\}$

The functionality A reveals A on every query. Operational depth is undefined: the unfolding $A \ni A \ni A \ni \dots$ does not terminate at $\emptyset$. Cyclic by Definition 3. Not well-founded.

The Quine atom belongs to the ZFA-mode of VR-Sets but not to the ZFC-mode. This is the first example of an operationally legitimate but non-well-founded set; detailed discussion appears in Part IV §IV.7.

Example 6: the set $B = \{\emptyset, B\}$

The functionality B reveals $\emptyset$ and B . Cyclic (reveals itself among its elements). The unfolding chain $B \ni \emptyset$ terminates at the first step; the chain $B \ni B \ni B \ni \dots$ does not terminate. Since well-foundedness requires termination of **every** chain, B is not well-founded.

This example shows that cyclicity need not be trivial (as with the Quine atom): a set may contain both ordinary elements and a self-reference.

Example 7: a description that does not specify a set

$R = \text{"reveal } x \text{ if and only if } x \notin x\text{"}$ (Russell's definition as considered above). This description is contradictory — it requires the functionality to give a contradictory response to the query " $R \in R?$ ". R is not operationally describable. R is not a set.

This is a critically important example: it shows that not every syntactic record specifies a set. The closure principle yields "every describable functionality is a set," but *describable* is a substantive condition, not empty: contradictory descriptions are excluded.

II.5. Summary and transition to Part III

In this part we have introduced the formal apparatus on which the remainder of the work rests:

(1) **Four definitions:** of a set, of $\emptyset$, of kinds of sets by operability, of operational identity $\equiv$.

(2) **Three lemmas:** extensionality ($\equiv$ is built into Definition 4), uniqueness of $\emptyset$, operational depth.

(3) **The closure principle:** every operationally describable functionality is a set. This is the principal instrument of the constructions that follow.

(4) **Resolution of Russell's paradox — operationally, not syntactically:** paradoxical descriptions do not specify functionalities.

(5) **Seven characteristic examples** covering the principal foundational situations: $\emptyset$, finite ordinals, ω , the Quine atom, a partially cyclic set, a contradictory description.

What follows. Part III applies this apparatus to the axioms of classical ZF. Each ZF axiom (except extensionality and the existence of $\emptyset$, already built into Part II) becomes a **closure theorem** of the operational universe — that is, an assertion that a corresponding describable functionality is a set under the closure principle.

Part IV formally separates the two modes in which the closure principle is applied — the ZFC-mode (only to well-founded functionalities) and the ZFA-mode (to all describable ones) — and establishes theorems of completeness for each mode relative to the corresponding classical theory. The Quine atom and its analogues are legitimate objects of the ZFA-mode but not of the ZFC-mode.

Part III. The ZF Axioms as Closure Theorems

0. Introduction

This part of the preprint shows that the eight standard ZF axioms (together with the axiom of choice AC) are, in VR-Sets, not postulates of existence but closure theorems of the operational universe relative to specific describable constructions. Two of them — extensionality and the existence of the empty set — have already been proved as Lemmas 1 and 2 of Part II and are only recalled here. The remaining seven assertions (pairing, union, power, infinity, replacement, foundation, choice) are derived by a uniform pattern on the basis of the closure principle fixed in Part II.

Proof pattern. For each axiom:

(1) the classical ZF formulation is given; (2) the operational functionality F corresponding to the asserted set is specified; (3) the operational describability of F is verified — that the functionality gives a definite response to every query; (4) by the closure principle it is concluded that F is a set.

Summary of results of Part III. The eight ZF axioms together with AC receive the following status in VR-Sets:

	Axiom		**Status in VR-Sets**	
	Extensionality		Lemma 1 (consequence of Definition $\equiv$)	
	Empty set		Lemma 2 (consequence of Definition 2)	
	Pairing		Closure theorem (trivially)	
	Union		Closure theorem (with a remark on $\equiv$)	
	Power		Closure theorem; $\wp(\omega)$ is countable (divergence)	
	Infinity		Theorem via operational infinity	
	Replacement		A single theorem in place of a schema (simplification)	
	Foundation		Mode-dependent: theorem in ZFC, false in ZFA	
	Choice		Theorem on the countable operational universe	

Terminological reminder. All terms — operational set, membership functionality, operational depth, operational identity $\equiv$, the closure principle — were introduced in Part II and the Foundation Notes and are used here without redefinition. The sign $\equiv$ denotes the operational identity of sets; the sign of equality $=$ does not appear in this part as an independent symbol, except in classical ZF formulations.

Three critical points. Before the detailed proofs we highlight three places where the results of this part are especially important for what follows:

- $\wp(\omega)$ in VR-Sets is countable. The sharpest divergence from classical ZF; it shows up already here, before the separate chapter on uncountability (Part VI).
- *The replacement schema collapses into a single theorem.* A structural simplification: the closure principle covers what in ZF requires an axiom schema.
- *Foundation is a mode-dependent property.* The distinction between the ZFC-mode and the ZFA-mode is defined precisely through the satisfaction or non-satisfaction of foundation. In Part IV this becomes a technical definition.

III.1. Extensionality

Classical ZF formulation

$$\forall A \forall B (\forall x (x \in A \Leftrightarrow x \in B) \rightarrow A = B)$$

Status in VR-Sets

Assertion (Lemma 1 of Part II). If sets A and B have the same elements (up to operational identity $\equiv$), then $A \equiv B$.

Remark. This is not a closure theorem: extensionality is built into the very definition of operational identity (Definition 4). The condition "for every element $x \in A$ there exists $y \in B$ such that $x \equiv y$, and conversely" is precisely the condition of coincidence of functionalities, i.e. $\equiv$ by definition. The statement "sets with the same elements are distinct" contradicts the definition of identity and is inexpressible in VR-Sets.

Relation to ZF. In ZF extensionality is a separate axiom because in the classical approach equality is given independently of membership. In VR-Sets equality (more precisely, operational identity) is *defined through* membership, and hence extensionality is a tautology of the definition. This is the first (and gentlest) of the places where the apparatus of VR-Sets proves more compact: a single decision at the level of definitions removes the need for a separate axiom.

III.2. Empty set

Classical ZF formulation

$$\exists A \forall x (x \notin A)$$

Status in VR-Sets

Assertion (Lemma 2 of Part II). There exists exactly one set with an empty functionality; it is denoted $\emptyset$ and is given as a primitive by Definition 2.

Remark. The existence of $\emptyset$ is not a closure theorem: $\emptyset$ is the nullary base operation of the construction, not an entity for which any ontological primacy is asserted (the slogan: "nothing is — all is doing"). Uniqueness was proved in Lemma 2: if two functionalities

both reveal nothing, the condition of coincidence of functionalities in Definition 4 holds vacuously on both sides.

Relation to ZF. In ZF the existence of the empty set is postulated by a separate axiom and justified by the separation schema. In VR-Sets $\emptyset$ is not an ontological primitive but the nullary base operation; there is no need to postulate it — $\emptyset$ is given by the construction of the system, itself an operation. The empty functionality is neither a theorem nor an axiom but the base of the operational construction; no ontology underlies it.

III.3. Pairing

Classical ZF formulation

$$\forall a \forall b \exists A \forall x (x \in A \Leftrightarrow x \equiv a \vee x \equiv b)$$

Theorem (pairing)

For any sets a and b there exists a set, denoted $\{a, b\}$, whose functionality reveals exactly a and b (when $a \equiv b$ — a single element).

Proof

Functionality. Define $F\{a,b\}$ as follows: on the first query the functionality reveals a ; on the second — reveals b ; on all subsequent queries reveals nothing. If $a \equiv b$, the second step amounts to a repeated presentation of the same element, and by extensionality (Lemma 1) the functionality reveals a single element.

Describability. On every concrete query the response is defined: 1st query $\rightarrow a$; 2nd query $\rightarrow b$; n -th query for $n \geq 3 \rightarrow$ exhaustion. No indeterminacy or contradictory responses arise.

Conclusion. By the closure principle, $F\{a,b\}$ is a set. It is denoted $\{a, b\}$.

■

Remark: ordered pairs

The ordered pair $\langle a, b \rangle$ is **not** introduced as an object of VR-Sets. This is consistent with the position of VR-Numbers, where pairs are excluded as objects: "nothing is — all is doing" means that new entities are not produced by "binding" already-existing ones, only by describable functionalities.

If in the sequel (for instance, in relations and functions) order should be required, it is realised not by introducing pair-objects but through *the order of revelation of elements within a functionality*. Order becomes an internal property of the description of a functionality, not a separate structure. Technically this means that a function $f : A \rightarrow B$ in VR-Sets is an operational set whose functionality, on the query "what does f yield at a ?", returns the corresponding element of B , without the mediation of a set of pairs $\langle a, f(a) \rangle$.

III.4. Union

Classical ZF formulation

$$\forall A \exists B \forall x (x \in B \Leftrightarrow \exists C (C \in A \wedge x \in C))$$

Theorem (union)

For any set A there exists a set, denoted $\cup A$, whose functionality reveals exactly all elements of elements of A (without repetitions up to $\equiv$).

Proof

Functionality. Define $F\cup A$ as a nested procedure: for each element C revealed by the functionality A, reveal in turn all elements x revealed by the functionality C; elements already previously revealed (up to $\equiv$) are skipped.

Describability. The functionality A is describable (A is the given set); for each revealed C the functionality C is describable (C is a set, an element of A); the composition of describable functionalities is a describable procedure. On every concrete query $\cup A$ gives a definite response: either a new element, or exhaustion (when all elements of all $C \in A$ have already been revealed).

Conclusion. By the closure principle, $F\cup A$ is a set. It is denoted $\cup A$.

■

Subtlety: decidability of $\equiv$ when skipping repetitions

The formulation of the functionality used the condition "skipping repetitions up to $\equiv$." In general, operational identity $\equiv$ is not algorithmically decidable: for arbitrary infinite sets A and B the question "A $\equiv$ B?" reduces to a coinductive check of coincidence of functionalities and in the worst case fails to terminate in a finite number of steps.

This does **not** render the functionality $\cup A$ impossible: its operational description is correct — at each step the decision "yield an element or skip" depends on the true answer to the question "does it coincide with one of those already yielded?" However, realising this functionality by an *effective algorithm* is not always possible. This distinction is consistent with the general position of VR-Sets: the theory is operational but not necessarily effective throughout. Decidability of $\equiv$ is a separate question for separate classes of sets (for finite sets with finite operational depth $\equiv$ is decidable; for VR numbers $\equiv$ is decidable by construction; in general — not).

An alternative formulation without appeal to $\equiv$: $F\cup A$ reveals all elements of all $C \in A$ *with possible repetitions*, and extensionality (Lemma 1) guarantees that the resulting set coincides with the set without repetitions. In this form describability does not depend on the decidability of $\equiv$, and the proof becomes more rigorous. In the sequel we take this as the canonical formulation.

III.5. Power

Classical ZF formulation

$$\forall A \exists B \forall x (x \in B \Leftrightarrow x \subseteq A)$$

Theorem (power)

For any set A there exists a set, denoted $\wp(A)$, whose functionality reveals exactly all subsets of A — that is, all operationally describable functionalities whose revealed elements belong to A .

Proof

Functionality. Define $F\wp(A)$ as a procedure revealing in turn all subsets of A .

Case of finite A . Let A have operational depth n . There are 2^n subsets; each is given by a functionality "reveal such-and-such elements of A , the others not at all," which is trivially describable. The enumeration of 2^n subsets is finite and describable.

Case of infinite A (for instance $A = \omega$). A subset of ω is any operationally describable functionality revealing some elements of ω . The set of such describable functionalities is **countable**: each functionality is given by a finite description (an algorithm of finite length) over a finite alphabet, and the set of finite strings over a finite alphabet is countable. The enumeration of these functionalities is describable (for instance, by increasing length of description and lexicographically within a single length).

Describability. In both cases, on every concrete query $F\wp(A)$ gives a definite response: the n -th subset of A by the enumeration, or exhaustion (for finite A).

Conclusion. By the closure principle, $F\wp(A)$ is a set. It is denoted $\wp(A)$.

■

Important consequence: $\wp(\omega)$ in VR-Sets is countable

From the proof it follows that $\wp(\omega)$ in VR-Sets is a countable set. This is a sharp divergence from classical ZF, where $\wp(\omega)$ is uncountable by Cantor's diagonal argument. The divergence calls for a direct comment.

A clarification on the status of $\wp(\omega)$. The proof above establishes $\wp(\omega)$ as a set by the closure principle: the description "the operability revealing in turn all describable subsets of A " specifies a functionality, and by closure this functionality is a set. This is "existence as an object." It is a separate question whether the same enumeration is realisable as an effective operational procedure — whether there exists an algorithm that, on the n -th query, returns a correct description of the n -th subset. The answer is no: such an algorithm would have to decide, for an arbitrary description, whether it specifies a correct functionality, which is equivalent to the halting problem. $\wp(\omega)$ is therefore countable as an object but not enumerable as a procedure. This distinction — countable from without vs. enumerable from within — is developed in detail in Part VI §VI.2 and

§VI.3, where it is shown to be precisely what removes Cantor's diagonal in the operational universe.

Where Cantor's diagonal fails to apply. The classical diagonal argument constructs, from an enumeration of subsets $S_0, S_1, S_2, \dots$, a set $D = \{n : n \notin S_n\}$ differing from every S_n . For this argument the enumeration of the S_n must be complete, and the definition of D must specify an operational functionality.

In VR-Sets a complete enumeration of all describable subsets of ω is **operationally unattainable** — it is equivalent to solving the halting problem: to enumerate all algorithms correctly specifying a subset of ω one would have to separate terminating functionalities from non-terminating ones, which is undecidable. Accordingly, the diagonal D , defined with respect to "all describable subsets," does not specify a describable functionality — it rests on an operational procedure which does not exist. By the closure principle D is not a set.

The diagonal is thus removed *not by forbidding* it as a proof, but by the fact that its premise (a complete enumeration of describable subsets) is operationally unrealisable.

Connection with computable analysis. An analogous situation is well known for computable real numbers: the set of computable reals is countable "from without" (from the standpoint of classical mathematics) but not enumerable "from within" — there is no algorithm enumerating all computable reals. $\wp(\omega)$ in VR-Sets stands in the same position: countable as an object, not enumerable as a functionality.

A detailed discussion of the consequences appears in Part VI. Here let us fix: the power set exists as a closure theorem, but **$\wp(\omega)$ in VR-Sets and $\wp(\omega)$ in classical ZF are not one and the same object**. The difference is most visible precisely at this axiom.

III.6. Infinity

Classical ZF formulation

$$\exists A (\emptyset \in A \wedge \forall x (x \in A \rightarrow x \cup \{x\} \in A))$$

Theorem (infinity)

There exists a set, denoted ω , whose functionality reveals in turn $\emptyset, t(\emptyset), t(t(\emptyset)), \dots$ without exhaustion and without cycling.

Proof

Functionality. Define F_ω : on the n -th query (for $n \geq 0$) the functionality reveals the n -th von Neumann ordinal O_n , where $O_0 = \emptyset$ and $O_{n+1} = t(O_n)$ (definitions from Part I of VR).

Describability. Each step of revelation is defined: the operator t (successor) is total on the objects of VR (axiom A3); by induction (axiom A4) the functionality is defined for every n . On every concrete query F_ω gives a definite response — a specific object O_n . The

functionality does not terminate (is not exhausted at any finite step) and is non-cyclic ($t(x) \neq x$ in VR — Theorem 5 of Part I).

Conclusion. By the closure principle, $F\omega$ is a set. It is denoted ω .

■

Remark: operational infinity

This statement lifts to the set level the principle of *operational infinity* from VR-Numbers (Part VI.3). In classical ZF the axiom of infinity postulates the *existence of a completed totality* of all natural numbers; finite means cannot derive its existence, so it is required as an axiom.

In VR-Sets no "completed totality" is asserted. ω is an **unboundedly applicable operation** enumerating O_n , and by the closure principle an unboundedly applicable describable operation is already a set. The divergence with classical ZF at this point is methodological: where ZF postulates, VR-Sets discovers a consequence of the operational universe.

ω has *non-finite* operational depth (unfolding to $\emptyset$ does not terminate in any finite number of steps), but is **non-cyclic**: the sequence $O_0, O_1, O_2, \dots$ is non-repeating. This distinction "infinite by enumeration" vs. "cyclic" first becomes operative at ω .

III.7. Replacement

Classical ZF formulation (axiom schema)

For every formula $\varphi(x, y)$ specifying a functional relation ($\forall x \exists! y \varphi(x, y)$) and for every set A there exists a set $B = \{ y : \exists x \in A, \varphi(x, y) \}$.

In ZF this is an **axiom schema** — one axiom for each formula φ of the language of ZF.

Theorem (replacement)

For any set A and any operationally describable procedure F defined on the elements of A , there exists a set, denoted $\text{image}(F, A)$, whose functionality reveals exactly $F(x)$ for every $x \in A$.

Proof

Functionality. Define Fimage : for each element x revealed by the functionality A , reveal $F(x)$.

Describability. The functionality A is describable. The procedure F is describable by assumption (this is part of the hypotheses of the theorem). The composition of describable functionalities is a describable functionality. On every query Fimage gives a definite response: for the next revealed $x \in A$ the result is a specific object $F(x)$, or exhaustion if A is exhausted.

Conclusion. By the closure principle, Fimage is a set. It is denoted $\text{image}(F, A)$.

■

Structural simplification: schema $\rightarrow$ single theorem

In classical ZF replacement is a schema: one axiom for each formula φ . This is bound up with the fact that in ZF a "relation" is a syntactic object (a formula of the language), and for each formula a separate existence assertion is required.

In VR-Sets an "operationally describable procedure F" is a single operational notion encompassing all ways of specifying functional relations (including all formulas of the language of ZF translated into operational descriptions). Therefore **a single closure theorem with respect to describable transformations replaces the entire schema.**

This is a pattern typical of VR-Sets: what in ZF is dispersed over an infinite series of syntactic cases is, in VR-Sets, gathered into a single substantive assertion resting on operationality (the closure principle), not on syntax.

Comparison with Fraenkel's replacement. Replacement was added to ZF by Fraenkel and Skolem in order to construct sets such as $\{\omega, \omega+1, \omega+2, \dots\}$, unreachable by Zermelo's means. In VR-Sets such sets are obtained directly through the closure principle: the procedure "apply t k times to ω " is describable, and so the image of this procedure on ω is a set. No additional postulates are needed.

III.8. Foundation (Regularity)

Classical ZF formulation

$$\forall A (A \neq \emptyset \rightarrow \exists x \in A (x \cap A = \emptyset))$$

Equivalent formulation

There exist no infinitely descending chains of membership: there is no sequence $x_0 \ni x_1 \ni x_2 \ni \dots$ without termination.

Status in VR-Sets: mode-dependent

This is the first ZF axiom whose status in VR-Sets **depends on the choice of mode** (ZFC-mode vs. ZFA-mode). Unlike the preceding axioms — closure theorems holding unconditionally — foundation turns out to be a *characterising property* of one of the two modes.

Assertion 1 (foundation in the ZFC-mode)

In the ZFC-mode of VR-Sets, foundation holds: there exist no infinitely descending chains of membership.

Proof

In the ZFC-mode, self-references of a label to its own functionality are syntactically forbidden. We show by induction on operational depth (Lemma 3 of Part II) that every set in the ZFC-mode has finite depth or "infinite-but-foundationally-grounded-by- ω -enumeration" depth.

Base. $\emptyset$ has depth 0. There is no membership chain from $\emptyset$ ($\emptyset$ contains no elements).

Step. Suppose the assertion holds for all sets of depth $< n$. Let A have depth n . Any chain $A \ni x_1 \ni x_2 \ni \dots$ begins with $x_1 \in A$; the depth of x_1 is strictly less than n (by the definition of operational depth); by the inductive hypothesis the chain from x_1 terminates at $\emptyset$. Hence the original chain terminates.

Case of ω -depth. For sets such as ω whose depth "runs to infinity" (and is non-cyclic), a chain from ω has the form $\omega \ni O_n \ni O_{n-1} \ni \dots \ni O_0 = \emptyset$. Any specific choice of an element $x_1 \in \omega$ is a specific O_n with finite n , and the subsequent chain terminates at $\emptyset$. An infinitely descending chain from ω does not arise.

■

Assertion 2 (failure of foundation in the ZFA-mode)

In the ZFA-mode foundation **does not** hold. There exist sets giving infinitely descending chains of membership.

Example

In the ZFA-mode the description of a set A is admissible through a label $A :=$ the functionality revealing one element — A itself. Formally: the functionality "on a query reveal the set denoted by the same label as the current one." The chain $A \ni A \ni A \ni \dots$ does not terminate. This set is known as a Quine atom in Aczel (1988).

Here it is important: A does **not contain itself** in the sense of physical containment. According to the operational world of VR-Sets (Foundation Notes §2), membership is a *reference*, not a container. $A \in A$ means that the functionality A returns A on a query — a specific definite response, not a structural self-embedding. There is no paradox; there is only a failure of foundation.

Conclusion

Foundation is not a universal theorem of VR-Sets but a **characterising property of the ZFC-mode**. The two modes differ precisely on this:

- **ZFC-mode** — foundation holds (the theorem above); self-references are syntactically forbidden; the operational depth of every set is defined.
- **ZFA-mode** — foundation fails; self-references are admitted; the operational depth of certain sets is undefined (cyclic functionalities).

In Part IV this will give a technical definition of the distinction between the two modes: ZFC-mode = ZFA-mode + foundation (when the restriction on self-references of labels is properly formulated). Conjecture: the ZFC-mode is equivalent to classical ZFC, the ZFA-mode to Aczel (AFA).

III.9. Choice

Classical ZF formulation (AC)

For every set of pairwise disjoint non-empty sets there exists a set containing exactly one element from each.

Status in VR-Sets: theorem on the countable universe

The position on AC is fixed in the Foundation Notes (§6.3) and is unfolded here into a proof. Briefly:

- **DC (dependent choice)** is built into the operational universe: every operation is carried out concretely, and a sequence of operations is built step by step. DC is not an axiom but a property of operability itself.
- **Full AC** is a theorem on the countable operational universe (for countable sets, AC reduces to DC).
- **The Banach–Tarski paradox** is not reproduced — not because AC is "operationally interpreted" but because there is no uncountable classical $\mathbb{R}$ in VR-Sets. Disputed applications of AC take place outside the operational universe.

Theorem (countable choice)

For every countable family $\{A_i\}_{i \in \omega}$ of pairwise disjoint non-empty sets there exists a set, denoted $\text{choice}(\{A_i\})$, containing exactly one element from each A_i .

Proof

Functionality. Define F_{choice} : on the i -th query the functionality reveals the first element of A_i — that is, the element revealed by the functionality A_i on its first query.

Describability. The countable family is an operational set whose functionality reveals $A_0, A_1, A_2, \dots$. Each A_i is an operational set with a non-empty functionality, so its first query gives a definite element. The composition of these describable procedures specifies a describable functionality F_{choice} : for every $i \in \omega$ a specific response is defined.

Conclusion. By the closure principle, F_{choice} is a set.

■

On full AC and the countability of the universe

The theorem just proved suffices for the entire operational universe of VR-Sets: the whole universe is countable (Foundation Notes §6.2), and so **any** family of non-empty pairwise disjoint sets in VR-Sets is countable. There are no uncountable families in VR-Sets. Accordingly, AC in full form is equivalent to countable AC, which is a theorem.

A detailed discussion appears in Part VI. Here let us fix: **AC is a theorem, not an axiom.** The Banach–Tarski paradox is absent not because AC is weakened, but because the objects on which it is built (uncountable subsets of $\mathbb{R}$, non-Lebesgue-measurable sets) do not exist in the operational universe.

Subtlety: the operability of "first element"

The formulation "reveal the first element of A_i " requires that the functionality A_i have a fixed "first" query. By Definition 1 of Part II the order of revelation of elements is an artefact of observation and does not enter into the identity of a set. However, for each concrete *presentation* of the functionality A_i its first query yields a definite element. If there are several presentations and the order of revelation differs, choice may return different elements — but these are different functionalities choice, not an inconsistent definition.

Formally: with a fixed order of revelation of each A_i , the function choice is uniquely defined. The operational theorem says that *such* a functionality is a set. It does not presuppose that there is a uniquely correct choice (which depends on the order of presentation); only that any consistent choice specifies a set. This is precisely the content of classical AC: existence of a choice function, not its uniqueness. The operational reformulation preserves this classical content exactly.

Conclusion

Part III shows that the ZF axioms in VR-Sets are not postulates but theorems. Of the nine assertions considered:

- **Two** (extensionality, empty set) are built into the definitions and lemmas of Part II and do not require a separate proof.
- **Six** (pairing, union, power, infinity, replacement, choice) are closure theorems of the operational universe with respect to specific describable functionalities. The closure principle gives them a uniform proof.
- **One** (foundation) is a mode-dependent property distinguishing the ZFC-mode from the ZFA-mode.

Substantive divergences from classical ZF brought out in this part:

(1) $\wp(\omega)$ in VR-Sets is countable. Cantor's diagonal argument in the operational universe produces no new subset, because its premise (a complete enumeration of describable

functionalities) is operationally unrealisable — it is equivalent to solving the halting problem.

(2) *The replacement schema reduces to a single theorem.* The closure principle with respect to describable transformations covers the entire series of cases that, in ZF, require a separate axiom for each formula.

(3) *Foundation is not a universal property but a distinguishing one.* The ZFC-mode and the ZFA-mode in VR-Sets are two operationally legitimate configurations; the distinction between them is a syntactic restriction on self-references of labels, technically manifesting as the satisfaction or non-satisfaction of foundation.

(4) *AC is a theorem, not an axiom.* On the countable operational universe AC reduces to DC, which is built into operationality. The Banach–Tarski paradox is not reproduced, not because of a weakening of AC but because of the absence of the classical $\mathbb{R}$ objects on which it is built.

What follows. Part IV uses foundation as the technical definition of the distinction between the ZFC-mode and the ZFA-mode, formulates this distinction syntactically, and discusses the conjecture of equivalence with classical ZFC and with Aczel. Part V specifies the role of VR numbers in VR-Sets: On are specific operational sets with finite depth, landing in the ZFC-mode as a theorem. Part VI develops the countability of the operational universe and discusses Cantor's diagonal, AC, and the absence of uncountability paradoxes in detail.

Part IV. The Two Modes: ZFC and ZFA

IV.1. The fork: two modes as two levels of application of the closure principle

In Part III the closure principle was applied to describable operational functionalities without restriction: every describable functionality is a set. In §III.8 it emerged that one of the ZF axioms — foundation — does not hold on all such functionalities. The Quine atom $A = \{A\}$ showed that the description "functionality revealing itself" is operationally describable but gives rise to a set with an infinitely descending chain of membership.

This is the foundational fork of VR-Sets. The closure principle in full strength gives a universe in which all ZF axioms hold **except foundation**. The closure principle, restricted by the condition that successive unfoldings terminate at $\emptyset$, gives a universe in which **all** ZFC axioms hold, including foundation. These two universes are the two modes of VR-Sets:

- **ZFC-mode**: the closure principle is applied only to functionalities whose successive unfoldings terminate at $\emptyset$.
- **ZFA-mode**: the closure principle is applied to all describable functionalities, without additional restriction.

In this part the modes are given a precise definition; it is proved that all ZFC axioms hold in the ZFC-mode (an assembly of the results of Part III); and the conjecture is discussed that the ZFC-mode is equivalent to classical ZFC and the ZFA-mode to the theory of Aczel (AFA, 1988). The corresponding converse directions — that every set of classical ZFC has an operational representation in the ZFC-mode of VR-Sets, and analogously for AFA — remain open; for each of them §IV.5 and §IV.6 analyse what is required for a proof.

Remark: the modes are not temporal

In VR-Sets there is no ontological primacy: $\emptyset$ is itself a nullary operation, and operability is all there is (not a self-standing being derived from a prior one, but action upon $\emptyset$ and upon the results of prior actions; see Part I §I.3 in detail). Considered as a description, however, operability is an integral object, not a process in time. When a functionality is given by its description, it **is** that description fully — there is no "moment of construction," no "already-built" or "being-built-just-now" in the operational world. Queries to a functionality are a way of addressing it, not moments of its being.

The distinction between the ZFC-mode and the ZFA-mode is therefore **not temporal** ("did the set manage to be built before reference was made to it?") but **structural** — a property of the functionality itself, considered as an integral object. The property is whether successive unfoldings of its responses terminate at $\emptyset$ in a finite number of steps.

This distinguishes the formulation of VR-Sets from the standard set-theoretic approach, where the well-foundedness condition is often described through an "iterative conception of the set" — the image of stepwise construction. In VR-Sets there are no steps and no construction; there is a structural property of a functionality.

IV.2. Unfolding, termination, operational depth

Before the formal definition of the modes we sharpen the technical apparatus inherited from Part II.

Unfolding of a functionality

Definition IV.1 (successive unfolding). Let A be an operational set. A *successive unfolding* of the functionality A is any sequence of sets

$$A = x_0 \ni x_1 \ni x_2 \ni x_3 \ni \dots$$

where x_{k+1} is one of the elements revealed by the functionality x_k . The unfolding *terminates at* $\emptyset$ if there exists n such that $x_n \equiv \emptyset$. The unfolding *does not terminate* if no such n exists.

The property of termination

Definition IV.2 (well-founded functionality). A functionality A is *well-founded* if **every** successive unfolding starting with A terminates at $\emptyset$ in a finite number of steps.

An equivalent formulation: there is no infinite sequence $x_0 \ni x_1 \ni x_2 \ni \dots$ starting with A . This is precisely the classical well-foundedness of the membership relation, lifted to an individual functionality.

Relation to operational depth

Lemma 3 of Part II defined the *operational depth* of a set as the least number of unfolding steps after which $\emptyset$ is reached. Cyclic sets (such as the Quine atom) have no finite depth because unfolding does not terminate.

Assertion IV.1. A functionality A is well-founded if and only if every set reachable from A by a finite chain of membership has a defined operational depth (finite, or ω -depth without cycling).

This is not a theorem in the strict sense but a development of definitions: well-foundedness is precisely the absence of cycles and the absence of infinite descending chains. For us it is important to fix that well-foundedness is a **structural property** of a functionality, verified on its description, not acquired at a moment of construction.

Examples

- $\emptyset$ is well-founded trivially (no elements, no unfoldings).

- Every VR number (On) is well-founded: its depth is n , and the single unfolding chain terminates at $\emptyset$ in n steps.
- ω is well-founded: any unfolding starting with ω yields on the first step some On with finite n , and then terminates in n steps. Despite the "infinity" of ω itself, each concrete unfolding is finite.
- The Quine atom $A = \{A\}$ is **not** well-founded: the unfolding $A \ni A \ni A \ni \dots$ does not terminate.
- The set $B = \{\emptyset, B\}$ (containing $\emptyset$ and itself) is not well-founded: the chain $B \ni B \ni B \ni \dots$ does not terminate, even though the chain $B \ni \emptyset$ does terminate.

The last example shows that well-foundedness is a property of **all** unfoldings, not of the existence of at least one terminating chain. This is in accord with the classical definition.

IV.3. Formal definitions of the ZFC-mode and the ZFA-mode

Having the notion of well-foundedness as a structural property of a functionality, we define the modes.

Definition IV.3 (ZFC-mode)

The **ZFC-mode** of VR-Sets is the operational universe in which the closure principle is applied only to well-founded operational functionalities. Formally: a set exists in the ZFC-mode if and only if (a) it is operationally describable and (b) it is well-founded.

Definition IV.4 (ZFA-mode)

The **ZFA-mode** of VR-Sets is the operational universe in which the closure principle is applied to all describable operational functionalities, without the well-foundedness restriction. Formally: a set exists in the ZFA-mode if and only if it is operationally describable.

Relation between the modes

Assertion IV.2. Every set of the ZFC-mode is a set of the ZFA-mode: $\text{ZFC-mode} \subseteq \text{ZFA-mode}$ as operational universes. The inclusion is strict: the Quine atom $A = \{A\}$ belongs to the ZFA-mode but not to the ZFC-mode.

The proof is trivial: the ZFC-mode is defined as the ZFA-mode with the additional condition of well-foundedness, so $\text{ZFC} \subseteq \text{ZFA}$. Strictness of the inclusion is witnessed by the Quine atom — it is describable (the functionality "reveal yourself" gives a definite response to every query) and not well-founded (the chain $A \ni A \ni \dots$ does not terminate).

■

Remark: the modes are not paradoxical

The Quine atom in the ZFA-mode is not paradoxical. Russell's paradox (Part II §5.2) is dissolved by the closure principle: the description $R = \text{"reveal } x \text{ if and only if } x \notin x\text{"}$ requires a contradictory response to the query " $R \in R?$ " and so does not specify a describable functionality. The Quine atom carries no such contradiction: "the functionality A reveals A " is a definite description, specifying a definite response to every query. That A **contains itself as an element** does not contradict the operational world of VR-Sets, because membership in VR-Sets is reference, not physical containment (Foundation Notes §2). $A \in A$ is "the functionality A in one of its responses returns A ," which is operationally defined.

The distinction between the ZFC-mode and the ZFA-mode is therefore not a distinction between "non-contradictory" and "paradoxical." Both modes are non-contradictory. The distinction lies in which operationalities are admitted.

IV.4. Theorem: all ZFC axioms hold in the ZFC-mode

This section assembles the results of Part III as they apply to the ZFC-mode and establishes a direct inclusion: the ZFC-mode of VR-Sets is a model of classical set theory ZFC.

Theorem IV.1 (completeness of ZFC in the ZFC-mode)

In the ZFC-mode of VR-Sets all ZFC axioms hold: extensionality, empty set, pairing, union, power, infinity, replacement, foundation, choice.

Proof

We show that for every ZFC axiom the corresponding closure theorem of Part III remains in force under restriction to well-founded sets — that is, that the constructions of Part III do not lead outside the ZFC-mode.

Extensionality (§III.1). Lemma 1 of Part II is formulated through $\equiv$ and applies to any sets, in particular to well-founded ones. The well-foundedness condition does not affect the condition of coincidence of functionalities.

Empty set (§III.2). $\emptyset$ is well-founded trivially (no elements means no unfoldings). $\emptyset$ belongs to the ZFC-mode.

Pairing (§III.3). Let a and b be well-founded. The functionality $\{a, b\}$ reveals a and b . Any unfolding starting with $\{a, b\}$ yields on the first step a or b — a well-founded set — and then the chain terminates by assumption. Hence $\{a, b\}$ is well-founded. The ZFC-mode is closed under pairing.

Union (§III.4). Let A be well-founded. $\cup A$ reveals elements of elements of A . Any unfolding of $\cup A$ yields on the first step some $x \in C$ for $C \in A$; the depth of x is strictly less

than the depth of A (two steps down membership); by the well-foundedness of A the chain terminates. The ZFC-mode is closed under union.

Power (§III.5). Let A be well-founded. $\wp(A)$ reveals all subsets of A — operationally describable functionalities revealing subsets of the elements of A . Each subset of A is well-founded (its elements are elements of A , well-founded by assumption; an unfolding one step down lands on a well-founded element of A). $\wp(A)$ is itself well-founded: its unfolding on the first step yields a well-founded subset of A . The ZFC-mode is closed under the power operation.

Infinity (§III.6). ω is well-founded: this was shown in §IV.2 (any unfolding of ω on the first step yields ω_n with finite n , and the chain terminates in n steps). ω belongs to the ZFC-mode.

Replacement (§III.7). Let A be well-founded and F a describable procedure mapping well-founded sets to well-founded sets. Then $\text{image}(F, A)$ consists of well-founded sets; the chain of unfolding $\text{image}(F, A)$ on the first step yields $F(x)$ — well-founded — and terminates by assumption. The ZFC-mode is closed under describable transformations preserving well-foundedness.

Foundation (§III.8). This is the defining property of the ZFC-mode: every set of the ZFC-mode is well-founded by Definition IV.3. Foundation holds automatically.

Choice (§III.9). Let $\{A_i\}_i \in \omega$ be a countable family of non-empty well-founded sets. The choice functionality reveals the first element of each A_i . The elements are well-founded (elements of well-founded sets are well-founded). The chain of unfolding choice on the first step yields a well-founded element — and terminates. The ZFC-mode is closed under countable choice, which on a countable operational universe yields full AC.

■

Corollary

Corollary IV.1. The ZFC-mode of VR-Sets is a model of the theory ZFC: every theorem of ZFC expressed in the language of operational sets holds in the ZFC-mode.

This is a direct consequence of Theorem IV.1 and of the fact that ZFC is precisely the theory derivable from these axioms. Substantively: the whole of standard mathematics formalisable in ZFC is accessible in the ZFC-mode of VR-Sets.

From this, however, it does **not** follow that the ZFC-mode is equivalent to ZFC. Equivalence requires the converse direction: that every model of ZFC is realised in the ZFC-mode of VR-Sets. This is an open question, treated in §IV.5.

IV.5. Conjecture: the ZFC-mode is equivalent to classical ZFC

The direct direction has been established: the ZFC-mode is a model of ZFC. The converse direction — that every model of ZFC is realised in the ZFC-mode of VR-Sets — is formulated as a conjecture, and we examine what is required for its proof.

Statement

Conjecture IV.1. The ZFC-mode of VR-Sets and classical ZFC are mutually interpretable: there exist operational interpretations of the standard model of ZFC in the ZFC-mode and conversely, agreeing on the membership relation.

What is already proved

The direct direction $ZFC\text{-mode} \rightarrow ZFC$ follows from Theorem IV.1: every statement provable in ZFC holds in the ZFC-mode, since all axioms hold. Accordingly, any operational construction correct in the ZFC-mode has an analogue in ZFC.

What is required for the converse direction

The converse direction $ZFC \rightarrow ZFC\text{-mode}$ requires that for every set of classical ZFC there exist an operationally describable representation in the ZFC-mode. This is stronger than "a model exists": classical ZFC proves the existence of sets for which an operational description is not obvious.

Principal obstacle: the uncountability of $\wp(\omega)$ in classical ZFC. In classical ZFC, $\wp(\omega)$ is uncountable (Cantor's theorem). In the ZFC-mode of VR-Sets, $\wp(\omega)$, as shown in §III.5, is countable: there are countably many operationally describable subsets of ω . This means that the classical $\wp(\omega)$ is **not** realised in the ZFC-mode as a single object.

Possible responses to this obstacle:

(a) *Conjecture in a weakened form.* To assert mutual interpretability not with respect to full classical ZFC but with respect to the *constructive part* of ZFC — the fragment yielding only operationally describable sets. This formulation is correct and provable, but less ambitious: it asserts the correspondence only for "constructive" mathematics.

(b) *Conjecture via submodels.* The ZFC-mode is interpreted as a countable transitive model of ZFC (or, more precisely, a fragment of one). By the Löwenheim–Skolem theorem such a model exists. The correspondence then is not between the ZFC-mode and the "real" classical ZFC, but between the ZFC-mode and one of the countable models of ZFC. This is consistent with the conjecture in the form "there is a countable model of ZFC realised in the ZFC-mode," and this direction appears the most promising for a rigorous proof.

(c) *Conjecture as a declaration of difference.* To acknowledge that the ZFC-mode of VR-Sets and full classical ZFC are **not** equivalent — they diverge precisely on those objects that classical ZFC proves to exist but which VR-Sets cannot specify operationally (such as

uncountable subsets of ω). In this formulation the conjecture becomes a theorem on divergence, and the work in §IV.5 becomes its formal fixation.

Our working position is (b): the ZFC-mode of VR-Sets corresponds to a **countable model** of classical ZFC. This position is technically realistic (countable models of ZFC are well studied), preserves the substantive connection with ZFC, and is consistent with the countability of the operational universe fixed in the Foundation Notes. A detailed proof of such a correspondence is a separate technical undertaking, beyond the scope of this preprint.

Methodological remark

The fact that the ZFC-mode of VR-Sets is not the "whole" of classical ZFC but a countable model of it is, in the VR system, not a defect but an expected consequence of the operational universe. "Nothing is — all is doing" means that VR admits only what can be specified operationally; operationally specifiable objects are finitely or countably many. The uncountable objects of classical ZFC are an artefact of the non-operational universe of classical mathematics, which admits the existence of objects without their operational description.

VR-Sets therefore does not claim to reproduce the full classical universe of sets. It claims to be an operationally coherent alternative in which all objects are operational and which nonetheless contains enough for the whole of constructive and countable mathematics.

IV.6. The ZFA-mode and Aczel

A parallel picture is built for the ZFA-mode. Aczel (1988) proposed the theory AFA (Anti-Foundation Axiom) — an extension of ZF in which foundation is replaced by anti-foundation: every directed graph is realised as the membership graph of a unique set. This permits the existence of cyclic sets such as the Quine atom.

Theorem IV.2 (completeness of ZFA in the ZFA-mode)

In the ZFA-mode of VR-Sets all ZFC axioms hold **except foundation**, and also the anti-foundation axiom AFA.

Proof

The axioms of extensionality, empty set, pairing, union, power, infinity, replacement, and choice are closure theorems of Part III. The proofs do not use well-foundedness and carry over to the ZFA-mode directly.

Foundation in the ZFA-mode **does not** hold: the Quine atom $A = \{A\}$ belongs to the ZFA-mode (a describable functionality) and violates foundation. This is in accord with the removal of the corresponding condition in Definition IV.4.

AFA in the ZFA-mode. Aczel's anti-foundation axiom asserts: for every directed graph G there exists a unique function assigning to each vertex a set so that the membership relation between these sets agrees with the edge relation of the graph. In the operational formulation: for every describable graph G (a graph whose vertices and edges can be operationally enumerated) there exists an operationally describable collection of functionalities realising the graph.

The describability of such a collection is a standard construction: every vertex v of the graph G specifies a functionality f_v which on a query reveals f_w for every edge $v \rightarrow w$. This is an operationally describable description (for describable G). By the closure principle in the ZFA-mode (without the well-foundedness condition) this is a collection of sets. Uniqueness up to bisimulation is a consequence of extensionality (Lemma 1 of Part II).

■

Remark: operational AFA vs. classical AFA

Classical AFA (Aczel 1988) ranges over *every* directed graph, including uncountable graphs without a finite description. The theorem above establishes AFA in the operational universe only for *describable* graphs — those whose vertices and edges admit a finite operational description. Call this *operational AFA*. This restriction is the exact parallel of the situation with classical ZFC and the ZFC-mode in §IV.5: the operational universe is countable, so its anti-foundation principle ranges only over countable describable graphs. For every classical undescribable graph there is no corresponding object in VR-Sets, just as there is no uncountable $\wp(\omega)$. The symmetry between the two modes is therefore: in the ZFC-mode the operational universe is a countable analogue of classical ZFC; in the ZFA-mode it is a countable analogue of classical ZFC + AFA. Conjecture IV.2 below asserts mutual interpretability via a countable model in this same sense.

Conjecture IV.2 (equivalence with AFA)

Conjecture. The ZFA-mode of VR-Sets and the theory ZFC + AFA are mutually interpretable (in the sense analogous to Conjecture IV.1: presumably via a countable model).

Obstacles and approaches parallel those of §IV.5. The direct direction *ZFA-mode* $\rightarrow$ *AFA* follows from Theorem IV.2. The converse direction faces the same difficulties with uncountability; the working solution is a correspondence with a countable model of AFA.

IV.7. The Quine atom and cyclic sets: examples in the ZFA-mode

To make concrete how the ZFA-mode works in practice, we examine several examples of non-well-founded sets and verify that they are operationally legitimate in VR-Sets.

Example IV.1: the Quine atom

Description: the functionality A reveals A on every query. Formally: $A = \{A\}$.

Operationality: on a query the functionality gives a response — A. This is a definite response; the functionality is operationally describable. By the closure principle (in the ZFA-mode) A is a set.

Well-foundedness: the chain $A \ni A \ni A \ni \dots$ does not terminate. A does **not** belong to the ZFC-mode.

Uniqueness: by extensionality there is exactly one set with this functionality. By AFA, every graph consisting of a single vertex with a self-loop is realised in a unique way — this is A.

Example IV.2: a pair of mutually referring sets

Description: sets B and C such that $B = \{C\}$ and $C = \{B\}$.

Operationality: the functionality B on a query reveals C. The functionality C on a query reveals B. Both descriptions are definite.

Unfolding chain: $B \ni C \ni B \ni C \ni \dots$ does not terminate. Neither B nor C belongs to the ZFC-mode.

Distinguishability: B and C are different sets (B contains C, C contains B; the functionalities do not coincide), but they are *bisimilar*: their membership graphs are isomorphic (a two-vertex cycle). In AFA, bisimilar sets are equal; in the ZFA-mode of VR-Sets, extensionality through $\equiv$ gives the same result if $\equiv$ is interpreted as bisimulation. This is consistent with the standard AFA literature.

Example IV.3: a set partially well-founded

Description: $D = \{\emptyset, D\}$. The functionality reveals $\emptyset$ and D itself.

Operationality: on the 1st query D reveals $\emptyset$; on the 2nd — D; on the 3rd, exhaustion (or D again, depending on the formulation). The description is definite.

Well-foundedness: the chain $D \ni \emptyset$ terminates at the first step. The chain $D \ni D \ni D \ni \dots$ does not terminate. Since **not every** chain terminates, D is not well-founded by Definition IV.2. D belongs to the ZFA-mode but not to the ZFC-mode.

This example shows that well-foundedness demands the termination of **all** chains, not the existence of at least one terminating chain. This is the standard formulation, here verified on concrete operational descriptions.

Example IV.4: an infinitely descending series

Description: a set E_0 such that $E_0 = \{E_1\}$, $E_1 = \{E_2\}$, $E_2 = \{E_3\}$, ..., and this sequence does not terminate.

The description of E_0 operationally requires the definition of all E_n — a countable series of functionalities. This is operationally describable provided there is a uniform description of the series (for instance, an indexed description of E_n by n).

Unfolding chain: $E_0 \ni E_1 \ni E_2 \ni \dots$ does not terminate. E_0 is not well-founded; it belongs to the ZFA-mode but not to the ZFC-mode.

The example shows that non-well-foundedness is not necessarily connected with self-reference: one can construct a descending series without a cycle. The ZFA-mode encompasses both kinds of such sets.

Remark: the ZFA-mode as a foundational option

All these sets are legitimate objects of the ZFA-mode. They carry no paradox (the closure principle has been checked on each description); they create no contradiction with the other axioms (Part III and Theorem IV.2 were proved without the well-foundedness condition).

The question of **which** mathematical applications such sets have is beyond the scope of this preprint. In the standard literature, AFA is applied to the semantics of circular phenomena: processes, recursive data structures, situational semantics (Barwise–Moss 1996), the theory of matchings with cyclic references. For VR-Sets it is important to fix: these applications are **operationally legitimate**, and the passage from the ZFC-mode to the ZFA-mode is not a departure from rigour but an explicit foundational choice.

IV.8. Relation to standard approaches

We briefly fix how the VR-Sets formulation of the two modes relates to standard approaches in set theory.

Aczel (1988)

Aczel introduced AFA as an anti-axiom: "every pointed graph has a unique realisation as a set." In VR-Sets this axiom is a theorem of the ZFA-mode (see §IV.6). The difference lies in the manner of formulation: in Aczel it is a postulate of existence; in VR-Sets a consequence of the closure principle applied to describable graphs.

Bisimulation in Aczel is the equivalence relation on sets identifying sets with isomorphic membership graphs. In VR-Sets, the operational identity $\equiv$ (Definition 4 of Part II) is structurally the same: coincidence of functionalities by elements, recursively up to $\equiv$. This gives a direct correspondence: $\equiv$ in VR-Sets = bisimulation in Aczel.

Barwise–Moss (1996)

Barwise and Moss in "Vicious Circles" developed in detail the applications of AFA to situational semantics and the theory of non-correspondence (non-well-founded sets for modelling self-reference in natural language). VR-Sets is compatible with these

applications: the ZFA-mode gives the same mathematical universe as standard AFA, up to Conjecture IV.2.

The iterative conception of the set

The standard picture of ZFC rests on the iterative conception: sets are built in stages of ordinals ($V_0 = \emptyset$, $V_{\alpha+1} = \wp(V_\alpha)$, with union at limit steps). This picture is *temporal*: a set exists from the stage at which it first appears.

In VR-Sets there is no such temporality (§IV.1). The ZFC-mode is not "built up in stages" but is defined by a structural property — well-foundedness. This is not a different theory; it is a different *manner* of describing the same (or, in view of §IV.5, the operationally accessible part of the) universe. The iterative conception and the operational one are two different methodological optics on a single mathematical structure.

Topoi and categorical set theory

In categorical set theory (Lawvere, ETCS, ETCC) the accent is shifted from "membership" to "functions." VR-Sets is closer to this tradition than to classical ZF: the membership functionality is a category of arrows "reveal an element," and $\equiv$ is an isomorphism in this category. The connection of VR-Sets with topoi is a separate substantive programme, indicated in Part IX as an open question.

Conclusion

In Part IV the fork between the ZFC-mode and the ZFA-mode has been formalised as two levels of application of the closure principle. The principal results:

- (1) Definitions IV.3 and IV.4 formulate the modes through a *structural* property of a functionality (well-foundedness — termination of all successive unfoldings at $\emptyset$), without introducing a temporal order of construction and without a separate layer of labels.
- (2) Theorem IV.1 establishes that the ZFC-mode is a model of classical ZFC: all nine axioms hold.
- (3) Theorem IV.2 establishes an analogous result for the ZFA-mode: all ZFC axioms hold except foundation, and AFA holds.
- (4) Conjectures IV.1 and IV.2 formulate equivalence with the classical theories, with an analysis of what is required for the converse directions (the working solution: correspondence via a countable model).
- (5) Examples IV.1–IV.4 make concrete the ZFA-mode by means of the Quine atom, cycles, partially well-founded sets, and infinitely descending series.

Methodological summing-up. The two modes are not two different "kinds" of sets and not two competing foundations, but two foundational options of a single system. The closure principle and the definitions of Part II operate in the same way in both modes.

The difference is the explicit choice of which operationalities to admit: only the well-founded ones (standard mathematics) or all the describable ones (mathematics with self-references and cycles).

This choice is made **once and explicitly**, not in each particular construction. After the mode is chosen, all work proceeds uniformly within it.

What follows. Part V specifies the role of VR numbers in VR-Sets: the von Neumann ordinals $\aleph_0$ and ω itself are specific examples of well-founded sets, landing in the ZFC-mode as a theorem. Arithmetic is inherited via direct isomorphism with VR Part I (no separate proofs are required). Part V is a short part fixing the correspondence and establishing that the numbers of VR in VR-Sets are precisely the numbers of classical arithmetic, presented operationally.

Part V. VR Numbers in VR-Sets

V.0. Introduction

This part of the preprint establishes the correspondence between the VR numbers — the von Neumann ordinals O_n , obtained in the VR system (Reznik, 2026) — and the operational sets of VR-Sets. The aim is modest: to show that the VR numbers are specific operational sets; that they fall automatically into the ZFC-mode as a theorem (not as a restriction); and that the arithmetic developed in VR (Part I §7) is inherited in VR-Sets via direct isomorphism, without the need for re-proof.

The part is short by design. The substantive work on the axiomatisation of arithmetic has already been done in VR; the present part only specifies how this arithmetic looks in the new operational world. No new arithmetic theorems are proved here.

Status reminder. Throughout this part the phrase “belongs to the ZFC-mode” is used in the strict technical sense established in Part IV: a set is well-founded and operationally describable. This is Theorem IV.1 territory and does not depend on Conjecture IV.1 (mutual interpretability of the ZFC-mode with a countable model of classical ZFC), which remains open. Accordingly, when this part says that VR numbers and the apparatus of VR-Numbers are inherited in VR-Sets without re-proof, the inheritance is from VR and from VR-Numbers themselves — via the operational isomorphism established below — not from classical ZFC.

V.1. VR numbers as operational sets

In VR (Part I, Definitions 4–5) the von Neumann ordinals are defined recursively:

$$O_0 := \emptyset, \quad O_{n+1} := t(O_n) = O_n \cup \{O_n\}.$$

Substantively:

$$O_0 = \emptyset,$$

$$O_1 = \{\emptyset\},$$

$$O_2 = \{\emptyset, \{\emptyset\}\},$$

$$O_3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\},$$

...

In VR-Sets each of these expressions is interpreted as the description of an operational functionality. Formally:

Definition V.1 (VR numbers in VR-Sets)

The operational set O_n is the functionality whose description is as follows: "reveal in turn $O_0, O_1, \dots, O_{n-1}$; on subsequent queries reveal nothing." Recursive base: $O_0 \equiv \emptyset$ (the empty operationality, revealing nothing).

Commentary. This is precisely what the formula $O_{n+1} = O_n \cup \{O_n\}$ says: the elements of O_{n+1} are all elements of O_n (that is, $O_0, \dots, O_{n-1}$) plus O_n itself. In the operational formulation this means that the functionality O_{n+1} reveals $O_0, \dots, O_{n-1}, O_n$ — $n+1$ elements in all.

The operational depth of O_n equals n (Lemma 3 of Part II): the unfolding $O_n \rightarrow O_{n-1} \rightarrow O_{n-2} \rightarrow \dots \rightarrow O_0 = \emptyset$ terminates in exactly n steps.

V.2. ω as an operational set

The set of natural numbers ω in VR-Sets is the functionality already introduced in Part II with the description:

Definition V.2 (ω)

ω is the operational set whose functionality, on the n -th query, reveals O_n . The functionality does not terminate at any finite step (it is infinite by Definition 3 of Part II) and is non-cyclic (ω is not among the revealed elements).

The existence of ω was proved in Part III §III.6 as a closure theorem: ω is operationally describable (on every query the response is defined — a specific O_n), and by the closure principle such a functionality is a set.

This accords with the axiom of infinity in its operational reformulation: ω in VR-Sets is not a "completed totality of all naturals" but an unboundedly applicable operation of enumerating O_n . The substantive divergence with classical ZF at this point is methodological, not mathematical: mathematically ω in VR-Sets is isomorphic to ω in classical ZF (see §V.4).

V.3. VR numbers in the ZFC-mode: theorem

The principal technical result of this part is that VR numbers fall automatically into the ZFC-mode of VR-Sets. This is not a restriction imposed on the construction, but a theorem about a structural property of the numbers.

Theorem V.1

Every VR number — O_n for arbitrary $n \geq 0$, as well as ω itself — is a well-founded operational set. Accordingly, all VR numbers belong to the ZFC-mode of VR-Sets.

Proof

Part 1: every O_n is well-founded. Induction on n .

Base ($n = 0$). $O_0 = \emptyset$ is well-founded trivially: $\emptyset$ has no elements, no unfoldings, and the termination requirement holds vacuously.

Step ($n \rightarrow n+1$). Suppose $O_0, O_1, \dots, O_n$ are all well-founded. We show that O_{n+1} is well-founded. Any successive unfolding starting with O_{n+1} yields on the first step one of the elements of O_{n+1} , that is, one of $O_0, \dots, O_n$. By the inductive hypothesis each of these is well-founded — and hence the chain of unfolding from that element terminates at $\emptyset$ in a finite number of steps. Consequently every unfolding of O_{n+1} terminates. O_{n+1} is well-founded.

By induction (axiom A4 from VR, lifted to the level of sets through Lemma 3 of Part II) all O_n are well-founded.

Part 2: ω is well-founded. Any successive unfolding starting with ω yields on the first step some element of ω — namely a specific O_n with finite n . By Part 1, O_n is well-founded; the unfolding chain from O_n terminates at $\emptyset$ in n steps. Consequently the chain of unfolding of ω terminates in $n+1$ steps. ω is well-founded.

Conclusion. By Definition IV.3 (Part IV), a set belongs to the ZFC-mode if it is operationally describable and well-founded. All O_n and ω are describable (Part III §III.6 for ω ; for O_n — by the construction of Definition V.1) and well-founded (Parts 1 and 2 of this proof). All VR numbers belong to the ZFC-mode.

■

Remark: correspondence with VR Part II.5

In VR (Part II §5) it was proved: $t(x) \neq x$ for every object of the system, and the chains of membership in the objects of VR are finite. These facts are reformulations of well-foundedness in the language of VR. Theorem V.1 is the same content, expressed in the language of VR-Sets:

- $t(x) \neq x$ in VR $\Leftrightarrow O_{n+1}$ is not cyclic on itself in VR-Sets (the unfolding of O_{n+1} yields elements of strictly smaller depth, not O_{n+1} itself).
- Chains of membership in VR are finite $\Leftrightarrow$ successive unfoldings in VR-Sets terminate at $\emptyset$.

No new proof was carried out; Theorem V.1 is a transcription of what has already been proved into the new language. This is consistent with the general methodological position: VR-Sets does not re-prove arithmetic, it interprets it.

V.4. Isomorphism with the natural numbers of VR

The VR numbers in VR-Sets are precisely the VR numbers of the original VR system, presented operationally. We fix this formally.

Theorem V.2 (isomorphism)

The mapping $\varphi: \mathbb{NVR} \rightarrow \mathbb{NVR}\text{-Sets}$, sending each VR number O_n of the original system to the corresponding operational functionality O_n in VR-Sets, is an isomorphism of structures:

- (a) *bijectivity*: to each VR number there corresponds exactly one operational functionality in VR-Sets, and conversely;
- (b) *preservation of succession*: $\varphi(t(On)) \equiv t(\varphi(On))$ in VR-Sets;
- (c) *preservation of membership*: $Om \in On$ in VR $\Leftrightarrow \varphi(Om) \in \varphi(On)$ in VR-Sets;
- (d) *preservation of arithmetic operations*: addition, multiplication, and exponentiation defined in VR Part I §7 agree with the corresponding operations on operational functionalities in VR-Sets.

Proof

- (a) Bijectivity by construction: φ is defined recursively through On in both systems; the recursion determines the correspondence uniquely.
- (b) Preservation of succession. In VR succession is given by axiom A3: $t(x) = x \cup \{x\}$. In VR-Sets, succession from On is the operational functionality revealing $O0, \dots, On-1, On$ (that is, everything that On reveals, plus On itself). This corresponds precisely to the union of the set of elements of On and $\{On\}$ — that is, $t(On) = On \cup \{On\}$ in VR.
- (c) Preservation of membership. In VR, $x \in y$ means that x is one of the objects mentioned in the construction of y by A3. In VR-Sets, $x \in y$ means that the functionality y reveals x as one of its elements. By the construction of Definition V.1 these two relations coincide.
- (d) Preservation of arithmetic operations. Addition, multiplication, and exponentiation in VR are defined recursively through t (Definitions 7–9 of Part I of VR). These definitions carry over into VR-Sets literally: they operate through succession and the base $O0$, and both constructions are coordinated between the systems by clauses (a), (b). All arithmetic theorems T1–T4 of VR Part I §7 (commutativity, associativity, distributivity, $O1 + O1 = O2$) are inherited in VR-Sets without re-proof.

■

Corollary V.1

The entire arithmetic developed in VR Part I — a Peano-equivalent theory of natural numbers with all standard operations and theorems — is available in VR-Sets through the isomorphism φ without the need for additional work.

Methodological commentary. This result is the consequence of a deliberate methodological decision: VR-Sets does not replace VR but interprets it. The axiomatic work justifying arithmetic was carried out in VR once; in VR-Sets, arithmetic appears as an already-prepared structure, presented operationally. This is consistent with the general logic of the cycle of works: VR provides the minimal axiomatic; VR-Numbers extends it to $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$; VR-Sets places all of this in a foundational context. No part of the cycle duplicates the work of another.

V.5. Extension to VR-Numbers

We briefly fix that not only the natural numbers, but the whole superstructure of VR-Numbers ($\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$) is inherited in VR-Sets through the same mechanism.

General principle

In VR-Numbers (Reznik, 2026) the numerical extensions are built as follows:

- $\mathbb{N}$ — the von Neumann ordinals On (operationally — the sole level with sets proper as objects);
- $\mathbb{Z}$, $\mathbb{Q}$ — operational syntax over $\mathbb{N}$ (not pair-objects, but operational procedures with order built into the description);
- $\mathbb{R}$ — algorithms computing real approximations (a countable subfield of the classical $\mathbb{R}$, isomorphic to the field of computable reals);
- $\mathbb{C}$ — a two-axis structure motivated by the duality of axiom A1 (the two implications $F \rightarrow F$ and $F \rightarrow \top$), with algebraic gluing through $i^2 = \ominus 1$.

All these extensions introduce no ontology: none adds anything beyond operations — $\emptyset$ itself is a nullary operation, not a primitive. Integers, rationals, reals, complex numbers are operational procedures of varying complexity over natural numbers, which are themselves operational sets.

Transition into VR-Sets

Since all extensions of VR-Numbers reduce to operational procedures over natural numbers, and since natural numbers are represented in VR-Sets through the isomorphism φ (Theorem V.2), the whole apparatus of VR-Numbers transfers into VR-Sets automatically:

- $\mathbb{Z}$, $\mathbb{Q}$ in VR-Sets are operational sets whose descriptions use elements of $\mathbb{N}$ VR-Sets as basic building material. All operational procedures of VR-Numbers (addition of integers via signed descriptions, rationals as numerator/denominator pairs given operationally, and so on) are applicable directly.
- $\mathbb{R}$ VR-Sets is a countable subset of the ZFC-mode containing operational sets representing computable algorithms. Each such set is well-founded (the elements of algorithms — rational approximations — are well-founded; the description of an algorithm is finite, not cycled on itself).
- $\mathbb{C}$ VR-Sets is the two-axis structure over $\mathbb{R}$ VR-Sets, preserving the algebraic relation $i^2 = \ominus 1$.

No new theorems are proved here. The aim of this section is to fix that the transition VR-Numbers $\rightarrow$ VR-Sets is as seamless as the transition VR $\rightarrow$ VR-Sets. A detailed operational description of each extension is given in VR-Numbers; here we only assert transferability.

V.6. Under the standard coding, VR numbers do not use the ZFA-mode

One structural fact deserves explicit mention: under the standard von Neumann coding of numbers used in this preprint, the VR numbers fall entirely within the ZFC-mode. The ZFA-mode — self-references, cycles, non-well-founded sets — is neither needed nor used for arithmetic in this coding. Alternative non-von-Neumann codings (including coinductive constructions in the ZFA-mode) are possible in principle and are noted as an open direction in Part IX, Extension 4; this assertion concerns only the standard coding.

Assertion V.1

No VR number — neither natural ($\mathbb{N}$ or ω), nor integer, nor rational, nor real, nor complex — is, under the standard coding used in this preprint, a cyclic or non-well-founded set.

Justification. For natural numbers and ω this is Theorem V.1. For $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$: the operational procedures defining these numbers in VR-Numbers operate through finite syntactic records over already-constructed natural numbers; none of these records refers to itself; consequently the describable functionalities are well-founded.

Methodological significance. This is consistent with the general programme of VR: classical mathematics — arithmetic, analysis, algebra — lives entirely in the ZFC-mode of VR-Sets. The ZFA-mode (the theory of non-correspondence, circular structures, situational semantics) is a separate area of application that does not intersect with numbers. The distinction between modes in VR-Sets is therefore not "different mathematics" but "different applications of a single foundation": classical mathematics of numbers uses the ZFC-mode as a theorem about a structural property of its objects; mathematics of self-referential structures uses the ZFA-mode as an explicit foundational choice.

V.7. Summary and transition to Part VI

This part has established the following:

- (1) **VR numbers are specific operational sets of VR-Sets.** Each $\mathbb{N}$ is a functionality with operational depth n ; ω is a functionality with non-finite but non-cyclic depth (Definitions V.1 and V.2).
- (2) **VR numbers automatically belong to the ZFC-mode** (Theorem V.1). This is not a restriction but a theorem about a structural property — the analogue of what in VR Part II.5 was expressed as $t(x) \neq x$ and the finiteness of chains of membership.
- (3) **Isomorphism with VR** (Theorem V.2): the VR numbers in VR-Sets are precisely the VR numbers of the original system, presented operationally. The whole arithmetic developed in VR is inherited without re-proof (Corollary V.1).

(4) **VR-Numbers transfers analogously:** $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are operational procedures over natural numbers, all well-founded, all in the ZFC-mode (§V.5).

(5) **Numbers do not use the ZFA-mode:** the whole classical mathematics of numbers lives in the ZFC-mode (§V.6).

What follows. Part VI unfolds the central divergence of VR-Sets with classical ZFC — the countability of the operational universe. Already in the present part it is visible that VR numbers live on a countable structure: $\mathbb{N}$ VR-Sets is countable, $\mathbb{R}$ VR-Sets is countable (isomorphic to the field of computable reals). Part VI establishes the countability of the entire operational universe in full detail, analyses Cantor's diagonal argument in the operational universe, proves AC as a theorem on the countable universe, and fixes the absence of the Banach–Tarski paradox by construction.

Part VI. Cardinality, Cantor, Choice

VI.0. Introduction

This part of the preprint is devoted to the most visible divergence of VR-Sets from classical ZFC: **the entire operational universe of VR-Sets is countable**. In classical ZF, already $\wp(\omega)$ is uncountable by Cantor's diagonal argument, and the universe V is a proper class of uncountable cardinality. In VR-Sets, $\wp(\omega)$ is countable (Part III §III.5), and this countability extends to the entire universe.

The divergence is not mathematical (the mathematics of countable and uncountable collections is identical in both systems under suitable formulation) but foundational: classical ZF admits the existence of objects without an operational description; VR-Sets admits only describable objects, and describable functionalities are countable in number.

In this part we:

- (1) precisely formulate and prove the theorem on the countability of the operational universe (§VI.1);
- (2) develop the analysis of Cantor's diagonal argument in the operational universe — where it fails and why (§VI.2);
- (3) discuss the distinction between "countable from without" and "not enumerable from within" by analogy with computable analysis (§VI.3);
- (4) prove AC as a theorem on the countable universe (§VI.4);
- (5) fix the absence of the Banach–Tarski paradox in VR-Sets as a consequence of the absence of non-measurable sets (§VI.5);
- (6) discuss the methodological significance of a countable operational universe — its relation to classical mathematics, constructivism, and computable analysis (§VI.6).

VI.1. Countability of the operational universe

Theorem VI.1

The collection of all operational sets of VR-Sets is countable.

Proof

Every operational set in VR-Sets is given by its description — a finite syntactic record of an operational procedure. Descriptions are written in a formal language: a fixed finite alphabet of symbols (the primitives $\emptyset, \rightarrow, t$ from VR, together with auxiliary symbols — brackets, commas, symbols for naming and referencing). A description is a finite string over this alphabet.

The set of all finite strings over a finite alphabet is countable: this is a basic fact of the theory of formal languages. Indeed, for each length n there are finitely many strings ($|\Sigma|^n$, where $|\Sigma|$ is the size of the alphabet), and a countable union of finite sets is countable.

Every operational set has a description (by the closure principle — Part II §II.3 — sets are precisely the describable functionalities). Consequently, the mapping "set $\rightarrow$ its description" is an injection from the collection of sets into the set of finite strings. The image of this injection is a subset of a countable set, hence itself countable.

The collection of operational sets of VR-Sets is countable.

■

Remarks on the theorem

(1) The description is not unique. The same set may be given by different descriptions. For instance, $\emptyset$ may be described as "the empty operationality," as "the set revealing nothing," as $O0$, and so on. This means that the mapping "set $\rightarrow$ description" is many-to-one, but the injection in the other direction — "description $\rightarrow$ the set it specifies" — is well-defined (up to $\equiv$). The collection of sets is the image of a countable collection of descriptions modulo extensionality $\equiv$, and the image of a countable set is countable.

(2) Descriptions are countable even for infinite sets. ω is infinite in operational depth, but the description of ω is finite: "the n -th query yields O_n ." $\mathbb{R}$ VR-Sets contains infinitely many reals, but each of them is given by a finite algorithm. The finiteness of the description is not in contradiction with the infinity of what it describes; the description is a syntactic record of a rule, and a rule can generate infinitely many consequences.

(3) Comparison with classical ZF. In classical ZF, the sentences of the language are also countably many (Löwenheim–Skolem), but the *objects* in the standard model are uncountable — because the classical theory admits the existence of objects not specified by any sentence. VR-Sets has no such gap between sentences and objects: each object is a describable functionality, and this relation is constitutive, not a property that objects may or may not have.

VI.2. Cantor's diagonal argument in the operational universe

The most famous proof of uncountability is Cantor's diagonal argument for $\wp(\omega)$ and for $\mathbb{R}$. In classical ZF, it constructs from any proposed enumeration of subsets of ω a new subset distinct from all those enumerated, and concludes that no such enumeration is complete.

In VR-Sets $\wp(\omega)$ is countable (Part III §III.5). This means that the diagonal argument in the operational universe **does not produce a new set**. We analyse precisely where it ceases to operate.

The classical argument

Suppose we have an enumeration $S_0, S_1, S_2, \dots$ of all subsets of ω . Define:

$$D = \{ n \in \omega : n \notin S_n \}.$$

The set D differs from each S_n at the element n : D contains n if and only if S_n does not contain n . Hence $D \neq S_n$ for every n . D is a new subset of ω not appearing in the enumeration, contradicting the completeness of the enumeration. Cantor's conclusion: no complete enumeration exists, $\wp(\omega)$ is uncountable.

What the classical argument presupposes

The argument uses two premises:

(a) **Completeness of the enumeration:** $S_0, S_1, \dots$ covers all subsets of ω .

(b) **D is a well-defined set:** the condition " $n \notin S_n$ " specifies a subset of ω .

In classical ZF both premises are accepted without reservation. In the operational universe each of them requires verification.

Where the argument fails in VR-Sets

The obstacle is in premise (a), and it is deep.

A complete enumeration of describable subsets of ω is operationally unrealisable.

Every subset of ω in VR-Sets is an operationally describable functionality. The collection of such functionalities is countable (Theorem VI.1). However, *enumerating* them in an operationally describable sequence $S_0, S_1, \dots$ is a different task. A description of such an enumeration requires that for every n there exist an operational procedure returning the n -th subset on the list. This procedure must:

(i) enumerate all finite descriptions over the alphabet of operational procedures — this is describable;

(ii) **verify that each description specifies a correct functionality** — that is, terminates on every query, gives a definite response, is not self-contradictory. This is an **undecidable task**: the question of whether an arbitrary description is correct is equivalent to the halting problem.

Consequently, an *operationally describable* enumeration of all describable subsets of ω does not exist. Premise (a) of the diagonal argument is unrealisable in the operational universe.

What this implies for D

The definition $D = \{ n \in \omega : n \notin S_n \}$ requires access to S_n for each n — that is, it requires an operational enumeration. If no such enumeration exists, the description of D does not specify an operational functionality: on the query " $n \in D?$ " the procedure would have to refer to S_n , which is undefined by the operational enumeration.

By the closure principle (Part II §II.3), a description that does not specify an operationally describable functionality does not give rise to a set. **D is not a set in VR-Sets.** The diagonal argument in the operational universe is not a proof of uncountability but a demonstration that the attempt to construct a complete enumeration encounters the halting problem.

Relation to the classical result

Cantor's classical result remains true: in classical ZF, $\wp(\omega)$ is uncountable. VR-Sets does not refute Cantor — it works on a different foundational basis. The difference may be formulated thus:

- **In classical ZF**, the existence of subsets of ω not specified by any description is admitted. Such subsets "are" in the sense of classical ontology and form an additional uncountability beyond the describable.
- **In VR-Sets**, there are no indescribable subsets: a set is a description, full stop. Cantor's uncountability is an artefact of admitting indescribable objects; in the operational universe that admission is not made, and uncountability does not arise.

This is consistent with the known observation of Skolem (the Skolem paradox): every set theory in a countable language has a countable model, and "uncountability" within such a model is a relative notion. VR-Sets is not simply a countable model of ZFC (Part IV §IV.5), but an operationally coherent alternative in which the countability of the operational universe is the natural state, not paradoxical.

VI.3. Countable from without vs. not enumerable from within

A subtle distinction important for the understanding of the status of countability in VR-Sets. It is well known in computability theory and computable analysis; VR-Sets shares this distinction and relies on it.

Setting

A collection X is called **countable from without** if it is countable as an object of classical set theory — that is, there exists a bijection between X and $\mathbb{N}$ in classical ZF.

A collection X is called **enumerable from within** (recursively enumerable) if there exists an operational procedure that, as its n -th output, returns the n -th element of X .

These two notions **do not coincide**. There exist countable collections that are not recursively enumerable. A classical example from computability theory is the set of indices of terminating programs: it is countable (a subset of $\mathbb{N}$) but not recursively enumerable (by Rice's theorem).

Application to VR-Sets

The collection of all describable subsets of ω in VR-Sets:

- **Countable from without** (Theorem VI.1): descriptions are countable, sets are countable.

- **Not enumerable from within**: there is no operational procedure that, for every n , returns the n -th describable subset of ω .

This distinction dissolves the apparent tension: "if $\wp(\omega)$ is countable, why cannot one exhibit an enumeration?" The answer: countability does not imply enumerability. The set exists as a countable object; but a procedure that would list its elements does not exist.

Analogy with computable reals

The same situation is well known in computable analysis for the field of computable reals:

- The field of computable reals is countable (a subfield of the classical $\mathbb{R}$, consisting of numbers with an operational description).
- However it is not enumerable: the halting problem prevents the separation of terminating algorithms for computing reals from non-terminating ones.

$\mathbb{R}$ VR-Sets stands in the same position: a countable subfield of the classical $\mathbb{R}$, isomorphic to the field of computable reals, not enumerable from within. This is consistent with the position of VR-Numbers and confirms that VR-Sets and computable analysis work on the same operational basis — the one in which "operational" means "algorithmically specifiable."

Where this is especially important

The distinction "from without vs. from within" dissolves several apparent problems:

(a) *Cantor's diagonal* requires enumerability from within; countability from without is insufficient. Since $\wp(\omega)$ in VR-Sets is countable from without but not enumerable from within, the diagonal does not construct D .

(b) *Full AC on the entire universe* does not reduce to simple "choose one element from each set of a countable family" — an operational procedure of choice is required, and for an arbitrary family it may not be describable. See §VI.4 for details.

(c) *The Löwenheim–Skolem theorem* predicts the existence of countable models of ZFC from without; in VR-Sets such a model is countable also from within (though not its entire structure is enumerable from within). This is consistent with Conjecture IV.1.

VI.4. The axiom of choice as a theorem

Part III §III.9 has already established: countable AC is a theorem of VR-Sets, and full AC on the countable universe is equivalent to the countable one. Here we develop this

result, examine the connection with DC (dependent choice), and fix what distinguishes the VR-Sets position from the standard treatment of choice in ZFC.

Three levels of choice

In classical set theory three forms of the axiom of choice are distinguished:

- **Countable Choice (CC):** for a countable family of non-empty sets there exists a choice function. Equivalent to countable AC, without which many basic facts of analysis cannot be proved (for instance, that a countable union of countable sets is countable).
- **Dependent Choice (DC):** for every binary relation R on a non-empty set A such that for every $x \in A$ there exists y with xRy , there exists a sequence $x_0, x_1, x_2, \dots$ with $x_n R x_{n+1}$ for all n . Stronger than CC.
- **Full AC:** for an arbitrary family of non-empty sets there exists a choice function. Stronger than DC; the source of the Banach–Tarski paradox.

Status of each level in VR-Sets

DC is built into the operational universe. Every operational procedure consists of specific steps: at each step the procedure makes a definite choice, and subsequent steps depend on those already made. This is precisely the structure of DC. In the operational universe, DC is not an axiom but a property of operational action itself: it is impossible to conceive of an operational procedure that does not make dependent choices.

Technically: for a binary relation R given by a describable procedure, a sequence $x_0, x_1, x_2, \dots$ is constructed operationally — choosing at each step the first element satisfying R with the preceding one. "First" here refers to the order of enumeration given by the description of R . This sequence is itself operationally describable and, by the closure principle, is a set.

Countable AC is a theorem (§III.9). The proof in §III.9 showed: for a countable family of non-empty sets, the choice function is operationally describable (on the i -th query, reveal the first element of A_i). By the closure principle, this is a set.

Full AC is a theorem on the countable universe. Since the entire universe of VR-Sets is countable (Theorem VI.1), every "family of non-empty sets" is a countable family. Hence full AC reduces to countable AC, which is a theorem.

Theorem VI.2 (AC in VR-Sets)

In VR-Sets the axiom of choice in its full form is a theorem: for every family of non-empty sets there exists an operational choice function.

Proof. The collection of sets in VR-Sets is countable (Theorem VI.1). A family of non-empty sets is an operational set whose functionality reveals the elements of the family. By countability the whole family has a description of the form $A_0, A_1, A_2, \dots$ (a countable enumeration). The choice function: $\text{choice}(i) = \text{the first element of } A_i$ — is a describable procedure. By the closure principle, choice is a set.

■

Subtlety: the operationality of choice

The formulation "the choice function choice" in this proof depends on the *order of presentation* of the elements of A_i and A_{i+1} : "the first element of A_i " is the element revealed by the functionality A_i on its first query. Different presentations may give different "first elements," and then choice is a different functionality (formally — several different sets choice).

There is nothing paradoxical here: the operational theorem says that *some* choice function exists, not that it is unique. This is consistent with the classical situation: AC in ZFC asserts the existence of a choice function, not its uniqueness.

Where VR-Sets differs from ZFC on AC

Substantive difference. In classical ZFC, full AC is an independent axiom. From ZF without AC one can build models in which AC is false, and these models are mathematically substantive (for instance, models with the axiom of determinacy AD). In VR-Sets there is no such gap: AC holds automatically by virtue of the countability of the universe. To reject AC in VR-Sets would mean to reject countability, and countability is a consequence of operationality, not a separate postulate.

Methodological difference. In ZFC, AC is often regarded as "suspect" because of its non-constructive consequences such as Banach–Tarski. In VR-Sets, AC is harmless, because the objects on which ZFC builds paradoxical consequences (non-measurable subsets of $\mathbb{R}$, ultrafilters on $\mathbb{N}$) do not exist in the operational universe. See §VI.5.

VI.5. Absence of the Banach–Tarski paradox

The most famous "pathological" consequence of AC is the Banach–Tarski theorem: the unit ball in $\mathbb{R}^3$ can be partitioned into a finite number of pieces, and from these pieces two unit balls can be assembled. The proof makes essential use of full AC to construct non-measurable subsets of the sphere.

In VR-Sets the Banach–Tarski paradox **is absent**. We fix precisely why — because this is the most graphic point where the operational universe departs from the classical one not only formally but substantively.

Where Banach–Tarski requires AC

The Banach–Tarski proof consists of several steps; the key one uses AC to construct a **non-measurable Vitali subset** of the sphere. A Vitali set is a set of representatives of equivalence classes on the sphere under an appropriate relation. The existence of such a set requires the selection of one representative from an uncountable family of classes — which is an application of full AC for uncountable cardinality.

Once the non-measurable set has been constructed, the rest of the proof combines the free group of rotations of $\mathbb{R}^3$ with this set, obtaining a partition of the ball into a finite number of pieces and a reconstruction of two balls.

Why this does not happen in VR-Sets

Each of the steps relies on classical objects absent in VR-Sets:

(1) **Uncountable family of equivalence classes on the sphere.** In classical ZFC, the sphere S^2 is uncountable (as a subset of $\mathbb{R}^3$, classically uncountable), and a factor under a suitable relation gives an uncountable family. In VR-Sets there is no classically-uncountable sphere: S^2 VR-Sets is a countable set of describable points, and any factor remains countable. No uncountable family of equivalence classes arises.

(2) **Non-measurable Vitali set.** The Vitali construction requires the selection of one representative from an uncountable family of classes. In VR-Sets there are no such families; countable AC gives a choice function without any non-measurability.

(3) **Classical $\mathbb{R}^3$.** The geometric arguments of Banach–Tarski operate on classical $\mathbb{R}^3$ — an uncountable space with the full apparatus of Lebesgue measure and non-measurable sets. $\mathbb{R}$ VR-Sets is isomorphic to the field of computable reals and countable; $\mathbb{R}^3$ VR-Sets is also countable. On a countable set, measure behaves standardly (countably additive), and non-measurable subsets do not arise.

Assertion VI.1

The Banach–Tarski paradox is not reproduced in VR-Sets. This follows not from a weakening of AC (AC in VR-Sets is a theorem) but from the absence of classically-uncountable objects on which the paradox is built.

Methodological commentary

The Banach–Tarski paradox is often regarded as "evidence of the pathology" of full AC in classical mathematics. The standard response is that this is the price of working with the uncountable continuum and its subsets without operational restrictions.

VR-Sets gives a different response: the paradox arises not from AC, but from the attempt to combine AC with the admission of classically-uncountable objects lacking an operational description. If one admits only operational objects, AC is a harmless theorem, and there are no paradoxes. The difference is not in the mathematical apparatus, but in the foundational commitments.

This is consistent with the position of Solovay, Shelah, and others in work on models of $ZF + DC$: in such models AD is compatible with DC, there are no non-measurable sets, and analysis works without paradoxes. VR-Sets is the operational reformulation of this mathematical line: what is achieved in Solovay's models by restricting the universe (only definable sets) is, in VR-Sets, the natural state of the operational universe.

VI.6. Methodological significance of a countable operational universe

Closing the part, we discuss how the countability of the operational universe relates to classical mathematics and other alternative approaches. This is a methodological section and contains no new theorems.

What VR-Sets loses

Direct losses in comparison with classical ZFC:

- **Uncountable cardinalities:** $\aleph_1, \aleph_2, \dots$ and the whole theory of uncountable ordinals are absent in VR-Sets as separate levels. Cardinality in VR-Sets takes only two values: "finite" and "countable."
- **The continuum hypothesis:** the question "is the cardinality of $\mathbb{R}$ greater than the cardinality of $\mathbb{N}$?" does not arise in VR-Sets, because both cardinalities are countable.
- **Uncountable classical objects:** the set of all functions $\mathbb{N} \rightarrow \mathbb{N}$, the set of all subsets of $\mathbb{R}$, equivalence classes with an uncountable number of representatives — all such objects *in their classical form* are absent. Their operational analogues — describable functions, describable subsets — exist and are countable.

What VR-Sets retains

Direct retentions:

- **All of arithmetic and number theory** (Part V).
- **All of computable mathematics:** algorithms, recursive functions, formal languages, computability theory — all of this works with operational objects by construction.
- **Analysis on computable reals:** $\mathbb{R}$ VR-Sets supports integration, differentiation, power series — everything that does not require the existence of non-measurable sets or uncountable cardinalities.
- **Constructive and reverse mathematics:** systems such as RCA0, WKL0, ACA0 — all formalisable in the countable operational universe of VR-Sets.
- **Algebra and combinatorics on countable structures:** the theory of groups, rings, fields, graphs — all of this for countable carriers works without modification.

Relation to other approaches

VR-Sets shares its foundational position with several well-known approaches in the foundations of mathematics:

- **Computable analysis** (Weihrauch, Pour-El–Richards): mathematics on computable objects, including computable reals. The operational universe of VR-Sets is isomorphic to the basic structure of computable analysis.
- **Constructivism** (Brouwer, Bishop): mathematics on constructively given objects. VR-Sets is close to this position foundationally, but not necessarily methodologically (for

instance, classical logic in VR-Sets operates without restrictions on the law of excluded middle).

- **Predicativism** (Poincaré, Weyl, Feferman): mathematics on predicatively definable objects. VR-Sets is still more restrictive (only operationally describable objects), but substantively close.

- **Solovay, ZF + DC + AD**: models of ZF with dependent choice and the axiom of determinacy, without full AC and without non-measurable sets. VR-Sets gives an operational formulation of this mathematical position.

The differences of VR-Sets from each of these approaches lie in the manner of formulation. All the listed approaches construct countable or near-countable mathematics by *restricting* the classical universe. VR-Sets constructs it by *foundational choice*: countability is a consequence of the operational universe, not a restriction imposed on the classical one.

Concluding commentary

The countability of the operational universe of VR-Sets is not a defect relative to classical mathematics but a position consistent with the slogan "nothing is — all is doing." All objects of VR-Sets are actions, and operationally describable actions are countable in number. Classical mathematics not fitting this picture (uncountable cardinalities, non-measurable sets, paradoxes of choice) works on a different ontology — the one in which the existence of objects without an operational description is admitted.

Which of the two ontologies is "correct" is not a mathematical question. Both ontologies are internally non-contradictory; both are fruitful. VR-Sets is the explicit fixing of the operational universe, with all its consequences. Part VII will discuss how this position interacts with self-referential statements and Tarski's theorem on the undefinability of truth.

VI.7. Summary

In this part we have established:

- (1) **Theorem VI.1**: the collection of all operational sets of VR-Sets is countable. Proof — through an injection into the set of finite descriptions over a finite alphabet.
- (2) **Cantor's diagonal argument does not operate in VR-Sets** (§VI.2): its premise of a complete enumeration of describable subsets is operationally unrealisable (equivalent to the halting problem), and so the diagonal D is not a set.
- (3) **Countable from without $\neq$ enumerable from within** (§VI.3): $\wp(\omega)$ in VR-Sets is countable as an object but not enumerable by procedure. Analogy with computable reals.

(4) **Theorem VI.2** (§VI.4): full AC in VR-Sets is a theorem, a consequence of the countability of the universe. DC is built into operationality; countable AC was already proved in Part III §III.9.

(5) **Assertion VI.1** (§VI.5): the Banach–Tarski paradox is absent in VR-Sets, not because of a weakening of AC but because of the absence of classically-uncountable objects on which the paradox is built.

(6) **Relation to other approaches** (§VI.6): VR-Sets is foundationally close to computable analysis, constructivism, predicativism, and models of ZF + DC + AD; it differs in the manner of formulation — by foundational choice, not by restriction of the classical universe.

What follows. Part VII treats Tarski's theorem on the undefinability of truth in the light of the operational universe and connects it with the duality of axiom A1 from VR. This is the most philosophically loaded part of the preprint; the formulation is cautious — it reformulates, it does not resolve.

Part VII. Truth and Self-Reference

VII.0. Introduction

This part of the preprint considers Tarski's theorem on the undefinability of truth in the light of the operational universe of VR-Sets. The aim is not to "resolve" or "circumvent" Tarski (this would be impossible: his theorem is a formal result, true in any sufficiently rich formal system) but to examine how the operational universe reformulates the same situation and what subtleties this clarifies.

The part has two substantive directions:

(1) **Analysis of the liar paradox** from the standpoint of the closure principle: paradoxical self-referential statements do not specify definite operational predicates, in analogy with Russell's paradox not specifying a set (Part II §II.3).

(2) **A hierarchy of levels of truth through t**: the idea that applications of the successor operator t from VR can be interpreted as a passage to a new level of truth at which the truth of the previous level may be discussed. This accords with Tarski's approach to a hierarchy of metalanguages.

The formulation is cautious: VR-Sets *reformulates* the situation, it does not overturn Tarski's theorem. No claim of resolving classical paradoxes in the sense of "they are now gone" is being made.

VII.1. Tarski's theorem on the undefinability of truth

A brief reminder of the classical result.

Statement

Tarski's theorem (1933) asserts: for every sufficiently rich formal system T (containing arithmetic) the predicate "is true in T " is not expressible inside T . More precisely: there is no formula $\text{Tr}(x)$ in the language of T such that for every sentence ϕ of the language of T :

$$T \vdash \text{Tr}(\ulcorner \phi \urcorner) \leftrightarrow \phi$$

where $\ulcorner \phi \urcorner$ is the encoding (Gödel number) of the sentence ϕ . If such a predicate were expressible, one could construct a sentence L asserting " L is false," yielding the classical liar paradox.

Significance of the result

Tarski showed that truth for the language T must be formulated in a *metalanguage* — an extension of T containing expressive means absent in T itself. This gave rise to a hierarchy of languages: object language, metalanguage, meta-metalanguage, ... each with its own level of truth.

The result is universal: it applies to any formal system capable of encoding its own syntax. ZF, ZFC, PA, VR — all these systems are subject to Tarski's theorem. VR-Sets too.

VII.2. The liar paradox through the closure principle

Before discussing the hierarchy of truth, we trace how the liar paradox appears in the operational universe. This is an exercise parallel to the dissolution of Russell's paradox in Part II §II.3.

The classical liar paradox

The sentence $L = \text{"L is false"}$ generates a contradiction: if L is true, then L is false (by the content of L); if L is false, then L is true (since this is what it asserts). Neither truth value is consistent with L itself.

Analogy with Russell's paradox

Russell's paradox in VR-Sets is dissolved by the fact that the description $R = \text{"reveal } x \sqcap x \notin x \text{"}$ demands a contradictory response to the query " $R \in R$?" and thus does not specify an operational functionality. R *does not exist* as a set — not forbidden, but operationally impossible.

The liar paradox has the same structure. The description of the predicate "is true" applied to L demands a contradictory response: on the query "is L true?" the procedure must return truth if and only if L is false — that is, when the procedure must return falsity. Contradictory responses. The description does not specify an operationally describable truth predicate for L .

Assertion VII.1

The paradoxical self-referential sentence $L = \text{"L is false"}$ does not specify an operationally defined truth value. In the operational universe L has no truth value — not because L is "neutral" or "unknowable," but because the description of truth for L is internally contradictory.

Relation to Tarski

This does not contradict Tarski's theorem. Tarski speaks of the inexpressibility of the truth predicate *for the entire language* inside the language itself. Assertion VII.1 speaks of the inexpressibility of truth *for a specific paradoxical sentence* — a particular case. VR-Sets gives an operational interpretation: the truth predicate does not exist as an operationally describable functionality, and paradoxical sentences are the clearest witnesses to this.

For *non*-paradoxical sentences the truth predicate exists *locally*: for each specific non-self-referential ϕ it is operationally definable whether ϕ is true in a given model (for instance, by verification of the formula). What is inexpressible is the *single* truth

predicate for the entire language — because the attempt to specify it runs into paradoxical sentences.

VII.3. A hierarchy of levels of truth through the successor operator t

Tarski proposed a resolution of the truth problem through a hierarchy of languages: language L_0 , metalanguage L_1 (containing a truth predicate for L_0), meta-metalanguage L_2 (containing a truth predicate for L_1), and so on. Each L_{n+1} is an extension of L_n in which truth for L_n is expressible, but not for L_{n+1} itself.

In VR-Sets this hierarchy has a natural operational realisation through the successor operator t from VR.

A note on what t means here. The operator t is the same operation as in VR axiom A3 and as used in Part V to construct the von Neumann ordinals: $t(x) = x \cup \{x\}$. There is no second “ t ”. What differs in §VII.3 is only the domain to which the operator is applied: in VR and Part V the input is $\emptyset$ and its successors O_n , generating numbers; here the input is an operational set representing an object language, generating a metalanguage. The arithmetical and the metalinguistic readings of t are two applications of the same operation to different starting objects, not two homonymous operators.

The idea

The application of t to a set A produces a new set $t(A) = A \cup \{A\}$, in which A appears as one of the elements. Substantively: t passes from A to *the level at which A may be discussed as an object*. At this new level one may formulate statements about properties of A which are inexpressible at the level of A itself.

As applied to a language: let L_n be represented by the operational set of words (sentences) of length $\leq n$ or complexity $\leq n$. Then $t(L_n) = L_n \cup \{L_n\}$ is a set containing L_n as an object. At the level of $t(L_n)$ one may speak about L_n as a whole, in particular formulate a truth predicate for the sentences of L_n .

Correspondence with Tarski

Tarski: L_0 is the object language; L_1 is the metalanguage expressing truth for L_0 ; and so on.

VR-Sets: L_0 is an operational set of the object language; $t(L_0)$ is the next level containing L_0 as an object; $t^2(L_0)$ is another level; and so on. The sequence of t -applications is the operational realisation of the hierarchy of metalanguages.

This is not a proof and not a new theorem. It is the indication of a structural correspondence: t in VR (succession as the passage to a new object containing the previous one as an element) corresponds structurally to the passage from a language to a metalanguage in Tarski (the metalanguage contains the object language as an object of study).

Where the correspondence is exact

Several places where the parallel operates:

(1) **The new level contains the old:** $t(A) \supset A$ in VR; the metalanguage contains the object language as a fragment.

(2) **At the new level, properties of the old are expressible:** $t(A)$ contains A as an element, so one may formulate predicates of the form "A has property P"; in the metalanguage, truth for the object language is expressible.

(3) **The hierarchy is unbounded:** t may be applied any number of times, yielding O_n with arbitrarily large n ; Tarski's hierarchy of metalanguages is likewise infinite.

(4) **Each level has its own truth predicate:** at the level $t_n(L_0)$ truth for L_0 is expressible, but not truth for $t_n(L_0)$ itself. Parallel with Tarski: each L_{n+1} expresses truth for L_n , but not for itself.

Where the correspondence is inexact

The parallel is structural, not literal. There are essential differences:

- Tarski's hierarchy is a hierarchy of *languages* including syntax and semantics. The hierarchy of t is a hierarchy of *sets*, not carrying any independent semantic load until interpreted.
- For Tarski, each L_{n+1} *contains a formal truth predicate* Tr_n for L_n . In VR-Sets the operation t merely passes to a wrapping set; the *formal expression* of the truth predicate is a separate construction that must be built at each level.
- The infinity of the t -hierarchy in VR is operational infinity (Part I §I.3). For Tarski it is ordinary infinity of ordinals; there are variants with transfinite levels. VR-Sets is restricted to ω -levels of t (by the countability of the universe — Part VI), which is weaker than the classical transfinite hierarchy.

Assertion VII.2 (cautious formulation)

The hierarchy of applications of the successor operator t from VR provides an operational realisation of the idea of Tarski's hierarchy of metalanguages, restricted to ω -levels. This is not a "resolution" of the problem of the definability of truth but an operational interpretation of Tarski's existing resolution within a more minimalist frame.

VII.4. The duality of A1 and self-reference

In the VR system (Part I, axiom A1) the duality of implication is fixed: from F both values $\{F, \top\}$ are reachable through $\rightarrow$; from $\top$ — only $\top$. This asymmetry is the structural characteristic of two-valued classical logic, expressed through the primacy of F .

In VR-Numbers (Part V) the same duality motivates the two-axis structure of $\mathbb{C}$: the $F \rightarrow F$ axis (real) and the $F \rightarrow T$ axis (imaginary). In VR-Sets the duality of A1 provides the basis for one additional observation on self-reference.

Observation

The self-referential sentence $L = "L \text{ is false}"$ contains a *one-sided* dependence: the value of L is determined through the value of L . If one tries to represent L as an operational procedure, this procedure must first compute L in order to determine L . Circularity.

The duality of A1 gives two directions: $F \rightarrow F$ (trivial self-identity) and $F \rightarrow T$ (passage to T). In the case of the self-referentiality of L , circularity stalls at $F \rightarrow F$: the procedure refers to itself without passing to a new value.

This is not a proof; it is a speculative comment. A substantive connection between the duality of A1 and the paradoxes of self-reference is an open direction noted in Part IX as one of the promising questions for further work.

Alpha/Omega/Logos (digression)

In the working notes of the VR cycle (not included in the formal part of the preprints) a triadic personal apparatus has been explored — Alpha (inner core), Omega (reflexive observer), Logos (medium of thought). This scheme corresponds structurally to the logical triad: F (Alpha, the initial state), T (Omega, the result), $\rightarrow$ (Logos, the passage).

In the context of Part VII one may note: Tarski's hierarchy of truth is a hierarchy of *Omega*: each level of the metalanguage is the reflexive observer of the previous level. This accords with the fact that the very notion "truth for language L " is already an observation of language L from without.

This digression is not part of the formal apparatus of VR-Sets and has no theorem status. It records that the philosophical background of the VR programme naturally accords with Tarski's approach to self-reference through a hierarchy.

VII.5. Summary and transition to Part VIII

In this part:

- (1) Tarski's theorem on the undefinability of truth has been recalled (§VII.1).
- (2) It has been established (Assertion VII.1, §VII.2) that paradoxical self-referential sentences in the operational universe have no truth value, for the same reason that Russell's paradox does not specify a set: the description of truth is contradictory.
- (3) A parallel has been proposed between Tarski's hierarchy of metalanguages and the hierarchy of applications of the successor operator t from VR (Assertion VII.2, §VII.3). The parallel is structural, not literal; bounded to ω -levels by countability.

(4) A speculative comment on the connection of A1-duality and the paradoxes of self-reference has been fixed (§VII.4); substantive development is an open direction.

The part formulates a **cautious position**: VR-Sets does not overturn Tarski's theorem and does not resolve the paradoxes of self-reference; it gives an operational interpretation of the existing picture within an operational frame. This is interpretation, not alternative.

What follows. Part VIII gives the final substantive discussion of the position of VR-Sets. The slogan "nothing is — all is doing," already used repeatedly, is unfolded into a detailed grounding of the foundational commitments. A parallel to VR-Numbers Part VI, with the accent on the specifics of operational set theory. Briefly, also — the relation to other positions in the foundations of mathematics (Platonism, formalism, constructivism).

Part VIII. Ontological Position

VIII.0. Introduction

This part of the preprint develops in detail the position of VR-Sets. The substantive commitments accumulated through Parts I–VII are gathered into a single picture and grounded as a coherent whole. The relation to other well-known positions in the foundations of mathematics — Platonism, formalism, constructivism, structuralism — is also discussed.

Part VIII is the parallel of Part VI of the VR-Numbers preprint (Reznik, 2026), where analogous work was carried out for the operational foundations of numbers. The emphasis here is transferred to the specifics of operational set theory: what changes when "nothing is — all is doing" is applied to sets.

The part contains no formal theorems. It is a substantive and philosophical exposition of the position for which all the remaining work was undertaken.

VIII.1. "Nothing is — all is doing"

The slogan running through all three works of the cycle has three levels of meaning, each of which operates in the application to VR-Sets:

Level 1: $\emptyset$ is the base operation, not an ontological primitive

In VR-Sets there is no primary entity: $\emptyset$ is the base operation, the empty operationality. All other "things" are actions. This is a foundational decision, not the consequence of any theorem. It is motivated by minimalism: the fewer primitives, the simpler the foundation and the smaller the arbitrary commitments.

The choice of $\emptyset$ specifically as the primitive is likewise motivated. $\emptyset$ is the only entity that can be characterised *purely negatively* (reveals nothing), without requiring a prior content. Any other primitive object would require some "matter," some initial content whose origin would have to be explained. $\emptyset$ is free of this problem by construction.

Level 2: operationality is derived

Operationality — action upon $\emptyset$ and upon the results of prior actions — is not a self-standing kind of being. An operational set is its functionality, but this functionality is not a separate entity; it is a way of acting, related to $\emptyset$.

This distinction is principled (Part I §I.3). Were operationality a "second kind of entity," we would have a dual foundation: $\emptyset$ + operationality. A dual foundation is not minimalist — it has two primitive types that must somehow be related. A derived foundation resolves this difficulty: operationality "is not" in the proper sense; it "happens" — only its result, visible as a set, is.

A clarifying metaphor — corrected as it is drawn: one thinks the number 1 "is" and "to add 1" is a mere action, but in VR even 1 is an operation (the successor applied to $\emptyset$). So too in VR-Sets: $\emptyset$ is not a primary being but the base operation, and "functionality as object" is operation itself, not a being beneath the doing. There are only operations.

Level 3: "doing" as a fundamental modality

The third level is the most philosophical. The slogan asserts not only what is and what is not, but also *in what manner exists* everything that is not $\emptyset$. The manner is "doing," action, process. This is a modality of doing, distinct from the classical "being" (Parmenides) and "becoming" (Heraclitus).

In the classical ontology of sets the modality of being predominates: a set *is* with all its properties at the moment of consideration, like a Platonic idea. In VR-Sets the modality of action predominates: a set *acts*, it reveals elements when queried, and this activity is not a property of the set but its essence.

Connection with other philosophical positions: the closest parallel is Whitehead's process ontology (Whitehead, 1929), where the basic entities are events (actual occasions), not objects. VR-Sets is not process philosophy in the full sense (Whitehead's events are temporal; the operationalities of VR-Sets are not), but the methodological kinship is real: in both, action is prior to object.

VIII.2. The foundational commitments of VR-Sets

We collect into one list all the foundational commitments accumulated in the preprint. Each is a consciously accepted decision, not a mathematical necessity.

Commitment 1: $\emptyset$ is the base operation, not a primitive

All "objects" of VR-Sets are derived from $\emptyset$. No atoms, urelements, individuals, ordered pairs as objects, classes as new entities. Only $\emptyset$ and operationalities upon it.

Cost: standard constructions requiring pairs (relations, functions via graphs of pairs) are reformulated operationally (see Part III §III.3, Part V §V.5). Such reformulations sometimes lengthen technical proofs; but the foundational gain — the absence of additional primitives — outweighs this.

Commitment 2: only describable objects

All sets of VR-Sets are operationally describable functionalities. Indescribable objects — those without a finite description — are not realised in the operational world. This is the principal divergence from classical ZF, in which the existence of objects without a description is admitted (for instance, a typical element of an uncountable set).

Cost: the countability of the entire universe (Part VI). Uncountable cardinalities, typical uncountable constructions, and paradoxical consequences of full AC on uncountable collections are inaccessible. Standard analysis on $\mathbb{R}$ is restricted to computable reals.

Commitment 3: operationality is derived

Operational functionalities are not a self-standing kind of being alongside $\emptyset$ but derived entities — actions upon $\emptyset$. $\emptyset$ alone is the base operation; operationality is in the sense of describedness, not in the sense of any primary givenness in being.

Cost: attentiveness in formulation is required. To say "there exists a set A" in the full sense of being is incorrect: only $\emptyset$ exists, while A is a describable action upon $\emptyset$. This reformulation adds philosophical discipline but does not alter mathematical claims.

Commitment 4: membership is reference, not containment

$x \in y$ is read as "the functionality y reveals x in one of its responses," that is, as the reference of y to x. Not as "x is physically located inside y." This dissolves the intuitive paradoxes of self-reference (Part I §I.4): $A \in A$ is not "A is located inside itself" but "the functionality A returns A in one of its responses."

Cost: a rejection of the container metaphor of a set, to which classical thought is accustomed. Worth it: the container metaphor is the source of most intuitive difficulties in the foundations of set theory.

Commitment 5: two modes as an explicit foundational choice

The admission or prohibition of self-references (Quine atoms, cycles) is not a mathematical necessity but an foundational choice: ZFC-mode or ZFA-mode. Both modes are legitimate; the choice is made explicitly and depending on the goals of application (Part IV).

Cost: the illusion of a "uniquely correct" set theory is lost. This is not a cost but a clarity: different applications require different frames, and explicit acknowledgement of this is a methodological advantage.

Commitment 6: hierarchy of truth through applications of t

Truth for VR-Sets is not expressible inside VR-Sets in full generality (Tarski's theorem). The hierarchy of applications of the successor operator t gives levels at each of which truth for the previous is expressible, but not for the current level (Part VII).

Cost: there is no single "above-all-levels" truth predicate. This is a commitment shared with all sufficiently rich formal systems by Tarski's theorem; it is not specific to VR-Sets.

VIII.3. Relation to classical positions

We briefly discuss how VR-Sets relates to the principal positions in the philosophy of the foundations of mathematics.

Platonism

Platonism (Gödel, Quine, the early Goodman) asserts that mathematical objects exist independently of our knowledge of them; mathematical cognition is discovery, not construction. Cantorian set theory in its modern form is in large part shaped by Platonist intuitions: the entire hierarchy V is a "really existing" universe, and our theories describe it more or less adequately.

VR-Sets is categorically non-Platonist. There are no sets in VR-Sets existing independently of description. "Nothing is — all is doing" is the operational minimum, excluding the entire classical Platonic hierarchy. Mathematical objects not operationally describable are absent in the operational world of VR-Sets — not because we have "not discovered them," but because there is no place for them in the operational world.

Formalism

Formalism (Hilbert) asserts that mathematics is the manipulation of formal symbols without interpretation; the "meaning" of mathematical statements is their derivability in a formal system. Truth in formalism reduces to provability.

VR-Sets is closer to formalism than to Platonism, but does not reduce to it. Operational descriptions are not "symbols without meaning" but instructions for functionalities, with an interpretation through action. The difference: formalism regards mathematics as a game of symbols; VR-Sets, as a system of actions for which symbols are merely a way of recording. The action is real (as a process), even if expressed through a formal description.

Constructivism

Constructivism (Brouwer, Heyting, Bishop) asserts that mathematical objects must be constructively given; existence is construction. In its radical form (intuitionism), the law of excluded middle is rejected.

VR-Sets is close to constructivism: all objects must be operationally given (= constructively built through description). However, VR-Sets does not reject classical logic. The law of excluded middle may be used in VR-Sets, since operational definiteness is already built into the closure principle: if a functionality is operationally defined, it gives a definite response; "the third" is operationally absent.

The difference: intuitionism rejects classical logic ontologically (there is no objective truth, only justification); VR-Sets retains classical logic (objects are defined, truth is locally definable) but restricts its universe to describable objects.

Structuralism

Structuralism (Benacerraf, Shapiro, Resnik) asserts that mathematical objects are positions in structures, and structure matters, not the nature of objects. The natural

number 3 is not "a particular object" but "the third position in any structure of natural numbers."

VR-Sets is compatible with structuralism and in a sense refines it. Operational sets are defined through *their behaviour* (what they reveal upon queries), not through "nature." Isomorphic functionalities are precisely one and the same set by extensionality $\equiv$. This accords with the structuralist intuition: structure matters, not matter.

However, VR-Sets is a more definite position than structuralism: it does not merely say "matter does not matter" but specifies concretely that nothing is — all is doing; $\emptyset$ too is an operation. Structuralism leaves open the question "what then specifically is?"; VR-Sets answers: there is no "is" — only doing.

The position of VR-Sets as "operationalism"

A suitable name for the position of VR-Sets is operationalism: mathematical objects are operational actions, and $\emptyset$ — the base they act upon — is itself a nullary operation, not an ontological primitive. This is:

- **not Platonism** — VR-Sets contains no objects independent of description;
- **not pure formalism** — symbols are interpreted through actions, not left without meaning;
- **compatible with constructivism foundationally but without** abandoning classical logic;
- **a refinement of structuralism**, specifying that "structure" is a pattern of actions upon $\emptyset$.

This position does not claim to be "uniquely correct." It is a coherent alternative, with its own advantages (operational minimalism, the absence of paradoxes of uncountability) and limitations (a countable universe, the absence of classical ZFC in its full extent).

A reminder on the status. The discussion in this part has positioned VR-Sets relative to classical ZFC, classical AFA, and other foundations. The *technical* relation between the ZFC-mode and a countable model of classical ZFC (and parallel for the ZFA-mode and AFA) is the content of Conjectures IV.1 and IV.2 and remains open; this is recorded in §IX.1 as Questions 1 and 2. The philosophical positioning of VR-Sets as operationalism does not depend on the resolution of these conjectures: the position stands on its own commitments, whether or not the technical equivalence with a countable model of the classical theory is established.

VIII.4. Applicability: for what VR-Sets is suited

Closing the discussion, we fix which mathematical applications are natural for VR-Sets and which are not.

Naturally suited

- **All of computable mathematics:** algorithms, recursive functions, formal languages, complexity theory, programming.
- **Constructive mathematics** in a broad sense, including reverse mathematics and predicative analysis.
- **Computable analysis:** analysis on computable reals and other computable structures.
- **Discrete mathematics:** graph theory, combinatorics, algebra over countable carriers.
- **Logic and model theory** for countable structures; formal verification of programs.
- **Non-well-founded sets** (through the ZFA-mode): modelling of circular phenomena in semantics, theory of processes, situational semantics of natural language.

Not directly suited

- **Standard analysis** on the classical uncountable $\mathbb{R}$ with the full theory of Lebesgue measure. The analogue in VR-Sets is analysis on $\mathbb{R}$ VR-Sets of computable reals, which covers most practical applications but differs formally.
- **The theory of uncountable cardinalities:** $\aleph_1$, uncountable ordinals, the continuum hypothesis. In VR-Sets these notions have no direct analogue.
- **Parts of classical functional analysis** relying on uncountable bases, Zorn's lemma for uncountable indices, the existence of ultrafilters. Constructive analogues exist, but not all classical results carry over.
- **Paradoxical consequences of AC:** Banach–Tarski and the like. They are absent by construction (Part VI §VI.5), which is for most purposes an advantage, but if for some reason they are required, VR-Sets is not suited.

The substantive position

VR-Sets is suited for mathematics in which *every object may be exhibited* — operationally, algorithmically, through a description. This is the greater part of substantive mathematics, including practically everything used in informatics, engineering, physics. VR-Sets is not suited for mathematics making essential use of non-exhibitable objects — typical elements of uncountable collections, non-measurable sets, equivalence classes with an uncountable number of representatives. This second area of mathematics exists and is fruitful; VR-Sets does not propose an alternative to it.

Accordingly, the choice between VR-Sets and classical ZFC is not mathematical and not philosophical in the narrow sense, but *methodological*: what kind of mathematics do we intend to pursue? For operationally transparent, computable mathematics, VR-Sets provides a coherent and minimalist frame. For mathematics requiring classical uncountability, ZFC or its extensions are needed.

VIII.5. Summary

In this part:

- (1) The slogan "nothing is — all is doing" has been unfolded on three levels of meaning: $\emptyset$ as a nullary operation (not a primitive), operationality as all there is, "doing" as a fundamental modality (§VIII.1).
- (2) Six foundational commitments of VR-Sets have been fixed, with the cost of each indicated (§VIII.2).
- (3) The position of VR-Sets has been related to Platonism (incompatible), formalism (close, but not reducible), constructivism (foundationally close but retaining classical logic), structuralism (compatible, refines). A suitable name for the position is operationalism (§VIII.3).
- (4) The applicability of VR-Sets has been described: naturally suited for all of computable and constructive mathematics; not directly suited for the classical theory of uncountable cardinalities (§VIII.4).

Methodological summing-up. VR-Sets is not "better" than classical ZFC and not "worse." It is a different foundational option, with its own area of applicability and its own advantages. The principal advantage is operational transparency: each object is given explicitly, each existence is grounded through description, and no assumption beyond operationality is made. The principal cost is the countability of the universe, excluding part of classical mathematics.

What follows. Part IX, the final part of the preprint, fixes the open questions: technical (formalisation in Lean, proofs of the equivalence conjectures, machine verification) and substantive (connections with topoi, homotopy type theory, extensions to Morse–Kelley and NBG).

Part IX. Open Questions

IX.0. Introduction

The concluding part of the preprint fixes the open questions of the VR-Sets programme. They are divided into three groups:

- (1) **Technical questions** — tasks of rigorously proving what has been formulated in the preprint as conjectures, and tasks of machine formalisation.
- (2) **Substantive extensions** — directions for developing the theory beyond the present preprint: connections with topoi, HoTT, extensions to MK and NBG, non-von-Neumann constructions.
- (3) **Philosophical questions** — substantive positions that have not received final formulation in the preprint.

Each question is formulated as a direction, not as a task with a known method of solution. Some of these questions may turn out to be technically routine (for instance, machine formalisation of already-proved theorems); others may require essentially new work.

IX.1. Technical questions

Question 1: Proof of Conjecture IV.1

Conjecture IV.1 asserts: the ZFC-mode of VR-Sets and classical ZFC are mutually interpretable (via a countable model of ZFC).

The direct direction was established in Theorem IV.1: the ZFC-mode is a model of ZFC. The converse direction — interpreting a countable model of ZFC in the ZFC-mode — requires:

- (a) the choice of a suitable countable model of ZFC (by Löwenheim–Skolem such models exist; a concrete construction convenient for operational interpretation is needed);
- (b) the construction of an operational representation of each element of this model as a describable functionality;
- (c) verification that the membership relation in the model is preserved under the passage to VR-Sets.

This is technical work feasible by standard means of model theory, but not trivial. An alternative approach — through the theorem on the existence of the minimal model of ZFC (L); the elements of L are describable by construction, which simplifies the operational interpretation.

Question 2: Proof of Conjecture IV.2

Conjecture IV.2 is the parallel of IV.1 for the ZFA-mode and the theory $ZFC + AFA$ (Aczel 1988). Technically similar, with additional difficulty on the AFA side: classical AFA theory also has an uncountable model, and a correspondence with the ZFA-mode requires work with a countable AFA-model.

Difficulty: in the AFA literature, countable models are less well studied than countable models of ZFC. A possible route — through Aczel's theorem on the final coalgebra of the power-set functor; the final coalgebra in a countable category may yield a countable AFA-model.

Question 3: Formalisation in Lean / Coq / Agda

Machine formalisation of the preprint in one of the proof assistants is the natural next task. This would provide:

- (a) a formal check of all proofs of Part III and Part IV;
- (b) a precise formulation of the definitions of Part II in the language of dependent types;
- (c) the possibility of machine verification of Conjectures IV.1 and IV.2 (at least the direct directions).

Lean 4 appears the most suitable due to its mathematical library (mathlib) and convenient support for coinductive definitions (for $\equiv$). Alternatively, Coq with its own library for non-well-founded sets, or Agda with its strong support for sized types.

Question 4: Precise characterisation of describable functionalities

The closure principle rests on the notion of an "operationally describable functionality." In the preprint this notion is used substantively but does not receive a precise formal definition. Possible formalisations:

- (a) through Turing machines: a describable functionality is one computable by a suitable machine;
- (b) through λ -calculus: a describable functionality is one expressible by a λ -term;
- (c) through primitive recursion and general recursion: a describable functionality is one given by a recursive definition.

These three formalisations are classically equivalent (by the Church–Turing thesis) but yield different practical approaches to working with VR-Sets. The choice of one or another formalisation is a separate technical question.

Question 5: Connection of $\equiv$ with bisimulation in rigorous formulation

In Part IV §IV.8 it was asserted that the operational identity $\equiv$ from Definition 4 structurally coincides with Aczel's bisimulation. This assertion requires a rigorous

formulation: which sets/graphs are considered, in which category the correspondence operates, how this equivalence interacts with the ZFC and ZFA modes.

IX.2. Substantive extensions

Extension 1: Connection with topos theory

Categorical set theory (Lawvere's ETCS and its extensions) considers sets through the category **Set** with its universal properties. VR-Sets is closer to this tradition than to classical ZF: $\equiv$ operates as an isomorphism in the category of operational functionalities.

Open questions:

- (a) Is the VR-Sets universe (for the ZFC-mode) a topos? If so, what are its specific properties (for instance, does it contain a subobject classifier)?
- (b) What is the connection between the ZFC-mode and the category of sets in a suitable topos of computable functions?
- (c) Can the ZFA-mode be obtained as a topos with a suitable modification (without the well-foundedness condition on structural morphisms)?

These questions form a bridge to the powerful categorical apparatus; their development could significantly extend the expressive power of VR-Sets.

Extension 2: Homotopy type theory (HoTT)

In HoTT, equality is not a relation but a type (path). This is consistent with the operational universe: $\equiv$ in VR-Sets is not a static relation but potentially a structure (two operationalities are equal in a particular way, and there may be many such ways).

Possible directions:

- (a) Reformulate VR-Sets in the language of HoTT, where $\equiv$ becomes a path type.
- (b) Investigate whether this yields new objects — operational sets with a non-trivial identity structure.
- (c) Connect the hierarchy of t-applications with n-types of HoTT.

This is a substantively close direction, but it would require considerable work in translating VR-Sets into a type calculus.

Extension 3: Morse–Kelley and NBG

In classical set theory, beyond ZFC, extensions with proper classes are considered: NBG (von Neumann–Bernays–Gödel) and Morse–Kelley. In these theories, "collections" too large to be sets are admitted.

In VR-Sets, the entire universe is countable, and so "too large collections" do not arise in the classical sense. However, operational analogues are possible:

(a) **Describable families of functionalities** that are not sets through failure of operational describability. For instance, "all terminating algorithms computing reals" — this is not a set in VR-Sets, since the corresponding family is not enumerable.

(b) **The hierarchy of meta-levels through t**: each $tn(\omega)$ is a set, but "the entire hierarchy" is an unbounded process, not a separate object.

A substantive development of the operational analogue of class theory is an open direction.

Extension 4: Non-von-Neumann codings of numbers

In Part V, VR numbers are represented as von Neumann ordinals $On = \{O0, \dots, On-1\}$. This is a standard coding but not the only one. Zermelo proposed an alternative: $\{\emptyset\}$, $\{\{\emptyset\}\}$, $\{\{\{\emptyset\}\}\}$, ...

In VR-Sets more exotic operational codings of numbers are possible — for instance, through infinite non-composite functionalities, through coinductive structures in the ZFA-mode, or through binary representations. Each such coding is a distinct operational realisation of $\mathbb{N}$; all are isomorphic as algebraic structures but differ in operational characteristics (operational depth, describability, and so on).

Open question: is there, among these codings, one distinguished in VR-Sets by a structural property (for instance, minimising operational depth or complexity of description)?

Consistency with §V.6. Part V §V.6 asserts that, under the standard von Neumann coding, VR numbers do not use the ZFA-mode. The present extension does not contradict this: it considers alternative non-von-Neumann codings as an open direction. Coinductive ZFA-mode codings of numbers, if developed, would be a separate operational realisation of $\mathbb{N}$ alongside the standard one — not a revision of the standard coding.

IX.3. Philosophical questions

Question 1: The nature of "action"

The slogan "nothing is — all is doing" appeals to the notion of "action" as a fundamental modality of doing. In Part VIII this was indicated as a parallel to Whitehead's process ontology, but it was not developed in full.

Open questions:

(a) What does "action" mean in the operational world of VR-Sets, given that time does not enter it? Operational functionalities are integral objects (Part IV §IV.1), not temporal processes. Then "action" is what?

(b) Can "action" be more fundamental than "object"? If so, this requires serious philosophical work not carried out in the preprint.

(c) The relation to other ontologies of action (Heraclitus, Bergson, Whitehead, process theology) is an open philosophical direction.

Question 2: The status of Conjectures IV.1 and IV.2

Should Conjectures IV.1 and IV.2 turn out to be false — that is, should VR-Sets prove to be substantively **not** equivalent to any countable model of ZFC or AFA — this would be a substantive result. It would mean that the operational universe yields its own mathematical structure, irreducible to the classical. Open question: to what then does the ZFC-mode of VR-Sets correspond, if not to a countable model of ZFC?

This question is not technical but substantive: it bears on the extent to which VR-Sets is its "own" theory and the extent to which it is a reformulation of the known.

Question 3: Applicability to physics

VR-Numbers (Reznik 2026) discussed possible connections with physics — through $\mathbb{C}$ as the natural algebra of quantum mechanics. VR-Sets potentially deepens this connection: a countable operational universe may be a suitable frame for discussing quantum states as operational descriptions of observables.

Open direction: is the operational world of VR-Sets compatible with the structure of Hilbert space used in quantum mechanics? Hilbert space is classically uncountable; computable Hilbert space (as in computable analysis) is countable. If physics in reality works only with computable states, VR-Sets is a natural operational frame for quantum mechanics.

This is a speculative direction not developed in the preprint; it is mentioned here as one of the promising areas of application.

Question 4: Relation to the VR programme as a whole

The cycle of works — VR. A Formal System, VR-Numbers, VR-Sets — is built as an ascending sequence: from arithmetic through numerical extensions to set theory. Open question: what is the *next* work in the cycle?

Possible candidates:

- (a) **VR-Analysis** — operational analysis on $\mathbb{R}$ VR-Sets.
- (b) **VR-Categories** — a categorical reconception of the cycle, with connections to topoi and HoTT.
- (c) **VR-Physics** — an operational reformulation of the foundations of physics.
- (d) **VR-Logic** — an extension of the basic logic of VR (two-valued classical) to intuitionistic, modal, or linear operational logics.

The choice of the next work is a substantive question depending on which direction proves most fruitful.

IX.4. Conclusion

The VR-Sets preprint, consisting of nine parts, has set out an operational theory of sets in which $\emptyset$ is a nullary operation, not an ontological primitive — there are only operations. The principal results:

- (1) The operational foundation (Parts I, II, VIII): a set is an operational functionality; membership is reference, not containment; operationality is derivative action upon $\emptyset$, not a self-standing kind of being.
- (2) **The closure principle** (Part II §II.3): the operational replacement for the axioms of existence. Every describable functionality is a set; paradoxical descriptions do not specify functionalities.
- (3) **The ZF axioms as theorems** (Part III): all nine ZF axioms + AC have been derived through the closure principle. Three key divergences from classical ZF: $\wp(\omega)$ is countable; the replacement schema becomes a single theorem; foundation is a mode-dependent property.
- (4) **The two modes** (Part IV): the ZFC-mode (well-founded functionalities) and the ZFA-mode (all describable). The distinction is structural, not temporal. Conjectures of equivalence with the classical theories via a countable model.
- (5) **VR numbers in VR-Sets** (Part V): von Neumann ordinals as specific well-founded sets; arithmetic inherited without re-proof.
- (6) **Cardinality** (Part VI): countability of the entire operational universe; Cantor's diagonal does not operate; AC is a theorem; absence of Banach–Tarski.
- (7) **Truth and self-reference** (Part VII): a cautious reformulation of Tarski's hierarchy through the successor operator t . Not a resolution but an operational interpretation.

The position of VR-Sets in the spectrum of foundations of mathematics. VR-Sets is an operationalist position located between formalism and constructivism, closer to computable analysis and predicativism than to Platonism. It does not claim to replace classical ZFC, but provides an alternative operational frame for mathematics in which every object may be exhibited operationally.

The principal substantive achievement is the demonstration that operational minimalism ("nothing is — all is doing") applied to sets yields a coherent, technically workable, and philosophically transparent theory, encompassing the greater part of substantive mathematics and dissolving classical paradoxes (Russell, Banach–Tarski, partially the Liar) operationally rather than syntactically.

The open questions gathered in this part indicate directions for further work — technical (formalisation in Lean, proofs of conjectures, precise characterisation of describability), substantive (connection with topoi, HoTT, extensions), and philosophical (the nature of "action," the next work in the VR cycle). Each of these directions is fruitful as an independent programme.

Closing the preprint, we repeat the slogan of the cycle:

"Nothing is — all is doing."

Applied to sets: $\emptyset$ is the base operation, not a primary being; everything else is operational action, describable by finite procedures, unfolding into a countable, transparent, coherent mathematical universe. This universe is VR-Sets.

Part X. Lean 4 Formalisation of VR-Sets: Methodological Observations

§X.1 Overview

The formalisation of VR-Sets Parts II–V in Lean 4 (mathlib v4.29.1) was completed in thirteen stages. The base type `OSet := ZFSet` — mathlib's `Quotient PSet.setoid` — carries all nine ZFC axioms as theorems and serves as the operational set universe. The formalisation uses the same Opus–Sonnet review architecture as VR-Numbers (§VIII). The same axiom ceiling `[propext, Classical.choice, Quot.sound]` is maintained; four objects achieve the empty axiom profile (no axioms whatsoever), a tighter result than VR-Numbers, where no object was axiom-free.

The results divide into three categories, each new relative to VR-Numbers:

1. **Proved theorems:** all nine ZFC axioms, Theorem V.1 (well-foundedness), Theorem V.2 (VR-OSet isomorphism), Theorem IV.1 (ZFC-mode collector). 2. **Refuted claims:** classical Anti-Foundation Axiom (AFA) and the Quine atom specification, both provably false in mathlib's `PSet`. 3. **Open formulations:** Conjectures IV.1 and IV.2, recorded as `def : Prop` — a third Lean status distinct from `theorem` and `axiom`.

The sections below record methodological observations arising from the formalisation, grouped thematically. Each observation is stated concisely; the underlying Lean evidence (definitions, proofs, `#print axioms` output) is in the formalisation repository.

§X.2 Bisimulation and the Quotient Base

Observation A.1 (Bisimulation as definitional equality). The preprint defines operational identity (Definition 4) as a bisimulation: «for every element x of A there exists an element y of B with $x \equiv y$, and conversely». In Lean, this is `PSet.Equiv` — the extensional bisimulation on pre-sets (Mathlib ZFC.PSet). The quotient `ZFSet := Quotient PSet.setoid` promotes `PSet.Equiv` to Lean's propositional equality `Eq`. Setting `OSet := ZFSet` (as an `abbrev`) makes `a  $\equiv$  b` syntactically identical to `a = b : OSet`. The bisimulation does not require a separate proof step — it is *definitionally absorbed* into the quotient construction.

This is more economical than VR-Numbers, where each isomorphism (`IntVRIntIso`, `RatVRIntIso`, etc.) required explicit `forward`/`backward` maps with round-trip proofs. Here, the bisimulation is the quotient.

Observation A.2 (Duplicate elements via quotient). The preprint states (§II.2): «the union reveals elements without repetitions up to $\equiv$ ». In `ZFSet`, duplicate suppression does not require a separate predicate or a decidability hypothesis. The quotient `ZFSet`

identifies extensionally equivalent elements automatically; `ZFSet.sUnion`` (union) and `ZFSet.powerset`` (power set) return elements of `ZFSet``, so membership is already modulo `PSet.Equiv``. No `DecidableEq`` typeclass or analogous instance is needed at the union/power level, in contrast to settings where set equality is a separate predicate.

§X.3 Structural Boundaries

This section records five places where the Lean formalisation reveals a structural boundary between the operational universe of VR-Sets and mathlib's type-theoretic infrastructure. The boundaries are of two types: *Lean is wider* (the Lean object admits more than the operationally describable) and *Lean is narrower* (the Lean object pre-commits to a restriction not present in the preprint's general formulation).

Observation B.1 (First boundary: Power set). Stage 5 formalises `⊆(A)` via `ZFSet.powerset``. The preprint §III.5 restricts the power set to *operationally describable* subsets, yielding a countable result for countable `A`. In Lean, `ZFSet.powerset`` contains *all* subsets — including non-describable ones. For `A =  $\omega$` , the resulting set is uncountable. This is the first structural boundary of the formalisation: Lean's `⊆( $\omega$ )` is strictly wider than the preprint's operational `⊆( $\omega$ )`. The discrepancy is metatheoretic (the condition «operationally describable» is not Lean-expressible) and is documented in the Stage 5 source.

This boundary parallels the `ℝ_VR` inexpressibility in §VIII.6 of VR-Numbers v1.0.2: both involve a countability restriction that is meaningful in the preprint's operational universe but metatheoretic from Lean's perspective.

Observation B.2 (Second boundary: Replacement). `Theorem_III_7_Replacement`` wraps mathlib's `ZFSet.replacement``. The preprint §III.7 restricts replacement to *operationally definable* functions. `ZFSet.replacement`` takes a *Lean function* `F : ZFSet → ZFSet` — the full class of Lean functions, not just describable ones. The formalisation records the classical theorem (wider than the preprint), with the boundary documented in a comment. The axiom dependency is `Classical.allZFSetDefinable``, reflecting that «all Lean functions are considered definable» is precisely the non-operational assumption.

Observation B.3 (Third boundary: Choice). `Theorem_III_9_Choice`` is proved via `Classical.choice``. The preprint §III.9 argues that AC is a theorem of *countable* VR-Sets: in a countable universe a choice function can be constructed algorithmically. In Lean, `Classical.choice`` is a single axiom that applies universally — it does not specialise to countable sets and does not produce an explicit algorithm. The preprint's argument (countability → constructive choice) is metatheoretic. Lean records AC as an axiom; it cannot express the countability restriction that would make it a theorem.

Three boundaries in one direction: power set, replacement, choice — each place where operational describability is lost in the classical Lean universe.

Observation B.4 (Fourth boundary: Foundation, opposite direction).

`Theorem\_III\_8\_Foundation` holds unconditionally on `OSet = ZFSet` because `ZFSet` is well-founded by construction: `PSet` is an inductive type, so all elements are accessible. The preprint §IV presents Foundation as *mode-dependent*: it holds in ZFC-mode but not in ZFA-mode (where cyclic sets exist). In Lean, there is no ZFA-mode to speak of (see Observation B.5), so Foundation is a universal theorem — but this is a *pre-commitment*, not a generalisation. Lean's type theory pre-commits the entire universe to ZFC-mode; the modal distinction of the preprint surfaces here not as two parallel universes but as a single universe plus an explicit boundary marker (`AFA\_Refuted`, Stage 10) and a formulation of the opposite mode as a conjecture (`Conjecture\_IV\_2\_Statement`, Stage 11). The modal analysis is not foreclosed — it is repositioned. This is a boundary in the *opposite* direction from B.1–B.3: Lean is *narrower*, not wider.

Observation B.5 (Fifth boundary: ZFA — total absence, type-theoretic). The ZFA-mode of VR-Sets (§IV.5–§IV.7) requires a universe of non-well-founded sets where the Quine atom $A = \{A\}$ exists and AFA holds. A systematic search of all of mathlib4 for terms `AFA`, `AntiFoundation`, `non-well-founded`, `coinductive` (for sets), `NonWellFounded`, and `Quine` returned **zero results**. mathlib contains no AFA or coinductive set infrastructure.

More importantly, `PSet` is an *inductive* type in Lean 4:

```
inductive PSet : Type (u + 1)
  | mk (α : Type u) (A : α → PSet) : PSet
```

Lean's inductive types have a built-in well-founded recursion principle. The direct consequence is `PSet.mem\_irrefl : ∀ x : PSet, x ∉ x`. The Quine atom would require `Q ∈ Q`, which contradicts `PSet.mem\_irrefl`. Therefore the Quine atom is *provably impossible* in PSet — not merely hard to construct.

This yields the strongest result of the formalisation:

```
theorem quineAtom_impossible : ¬quineAtomSpec -- axiom-free proof
theorem AFA_Refuted : ¬AFA_Statement          -- axiom-free proof
```

Both are proved with *no axioms* (`#print axioms` returns `[]`). The proof of `AFA\_Refuted` applies AFA to the universal self-loop graph ($V = \text{Unit}$, $E_{_} = \text{True}$) and derives `f () ∈ f ()`, contradicting `PSet.mem\_irrefl`. No `propext`, no `Classical.choice`, no `Quot.sound`.

This boundary is categorically stronger than boundaries B.1–B.4: - At B.1–B.3, Lean's object *exists* (as a classical entity) and is merely *wider* than the operational version. - At B.4, the operational restriction is *absent*, but no inconsistency arises. - At B.5, the ZFA-mode universe *provably does not exist* in mathlib's type hierarchy. Adding AFA as an axiom would be inconsistent with `PSet.mem\_irrefl` already present in mathlib.

This is categorically stronger than the VR-Numbers §VIII.6 boundary at `ℝ\_VR`. There, classical `ℝ` existed in Lean and the computability restriction was *metatheoretic* — an

unexpressed condition on the type. Here, the ZFA-mode universe has *no representation at all* in mathlib's type hierarchy, and the impossibility is *constructively provable* (axiom-free proof). The boundary moves from "metatheoretic" to "structurally proven" — a stronger form of inexpressibility.

§X.4 Axiom-Minimal Patterns

Observation C.1 (Four faces of Classical.choice). ``Classical.choice`` enters the VR-Sets formalisation through four distinct mechanisms — none of which is the direct application of AC to a set-theoretic family:

```
| Theorem | Source of Classical.choice |
| `Lemma_II_3_DepthMono` (rank) | `Ordinal.iSup` (supremum of ordinal-valued function) |
| `Theorem_III_7_Replacement` | `Classical.allZFSetDefinable` (all Lean functions are «definable») |
| `Theorem_III_8_Foundation` | `WellFounded.has_min` (existence of a minimal element) |
| `Theorem_III_9_Choice` | `Classical.choice` directly (AC on ZFSet families) |
```

This parallels VR-Numbers, where the boundary into ``Classical.choice`` came primarily through a single structural place — mathlib's ``Rat.add`` normalisation via ``Nat.gcd`` (VR-Numbers §VIII.4). In VR-Sets, by contrast, the boundary is *distributed* across four mechanically distinct sources, reflecting the broader algebraic surface of set theory compared to number theory.

Observation C.2 (Axiom profiles of formulations reflect structural nature). The two conjectures of Stage 11 have different axiom profiles:

```
Conjecture_IV_1_Statement : [propext, Quot.sound]
```

```
Conjecture_IV_2_Statement : [] (no axioms)
```

``Conjecture_IV_1_Statement`` references ``OSet.{0} = ZFSet.{0}`` — a specific mathlib quotient type — so it inherits ``Quot.sound`` from the quotient construction and ``propext`` from membership on ``ZFSet``. ``Conjecture_IV_2_Statement`` is stated over abstract ``(U : Type, mem : U → U → Prop)`` — no reference to any specific mathlib type — and is therefore axiom-free.

This is not an incidental technical difference. It reflects what the conjectures *point to*: Conjecture IV.1 asks about a countable submodel of a specific mathlib type (`OSet`); Conjecture IV.2 asks about the existence of an abstract type with AFA structure — something *outside* the current mathlib type hierarchy. The axiom profiles witness this structural distinction at the level of dependency closures.

§X.5 Methodological Convergences

Observation D.1 (Replacement: schema $\rightarrow$ single theorem). In classical ZF, the Axiom of Replacement is a *schema*: one axiom instance for each first-order formula $\varphi(x, y)$ defining a functional relation. In the preprint §III.7, replacement is stated as a single theorem (over all operationally definable functions, where definability is built into the operational semantics). In Lean 4, `ZFSet.replacement` is a single polymorphic function: the schema is unified into one statement via function quantification. This is a three-way convergence: classical ZF (schema), operational VR-Sets (single theorem over describable functions), and Lean (single theorem over all functions) all agree that replacement is not a logical schema but a single closure principle — the classical version merely lacks a mechanism to unify function quantification.

Observation D.2 (Three-tier formalisation result). VR-Numbers produced only *positive* results: theorems proved, isomorphisms constructed. VR-Sets produces a structurally richer outcome:

Category	Objects	Example	
Proved theorems	16	<code>`Theorem_IV_1_ZFCAxioms`</code> , <code>`Theorem_V_2`</code>	
Refuted claims	2	<code>`AFA_Refuted`</code> , <code>`quineAtom_impossible`</code>	
Open formulations	2	<code>`Conjecture_IV_1_Statement`</code> , <code>`Conjecture_IV_2_Statement`</code>	

The refuted claims are *structural boundary theorems*: they arise because mathlib's type hierarchy cannot accommodate ZFA-mode, not because ZFA-mode is logically inconsistent. The open formulations are the first instances in the VR Lean cycle of a theorem-like object that the system records but cannot resolve.

This three-tier structure is a consequence of the modal analysis of Parts IV: any formalisation of a system that distinguishes modes (ZFC vs. ZFA) must confront the question of which mode the formalisation framework itself inhabits. Lean's type theory inhabits ZFC-mode by construction; VR-Sets is designed to be mode-agnostic. The boundary between these is the primary finding of Stage 10.

§X.6 Conclusion

The Lean 4 formalisation of VR-Sets demonstrates that the operational set theory of Parts II–V is machine-verifiable, with no axioms beyond the standard Lean ceiling `[propext, Classical.choice, Quot.sound]`. The formalisation required no new axioms, no ``sorry``, and no changes to mathlib.

The most significant finding — the axiom-free refutation of AFA and the Quine atom specification (Observation B.5) — was not anticipated in the formalisation plan. It arises

from a structural property of Lean 4's inductive type system (induction, not coinduction, for `PSet`) and represents the deepest boundary between the operational VR-Sets universe and classical Lean's type hierarchy.

The Lean formalisation generated seventeen methodological observations across thirteen stages; the ten presented in §X.2–§X.5 are those most directly bearing on the relationship between operational and type-theoretic foundations. The remaining seven (technical details of axiom inheritance, mathlib API quirks, and intermediate steps) are documented in the source comments of the formalisation. Together, they document not only what was proved, but what could not be proved, why, and what the corresponding structural boundary implies for the relationship between operational set theory and classical type-theoretic foundations.

Lean 4 formalisation repository: VRCycle, tag v1.2-vr-sets. Zenodo Software DOI: 10.5281/zenodo.20354340. Lean toolchain: v4.29.1 + mathlib v4.29.1.

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